



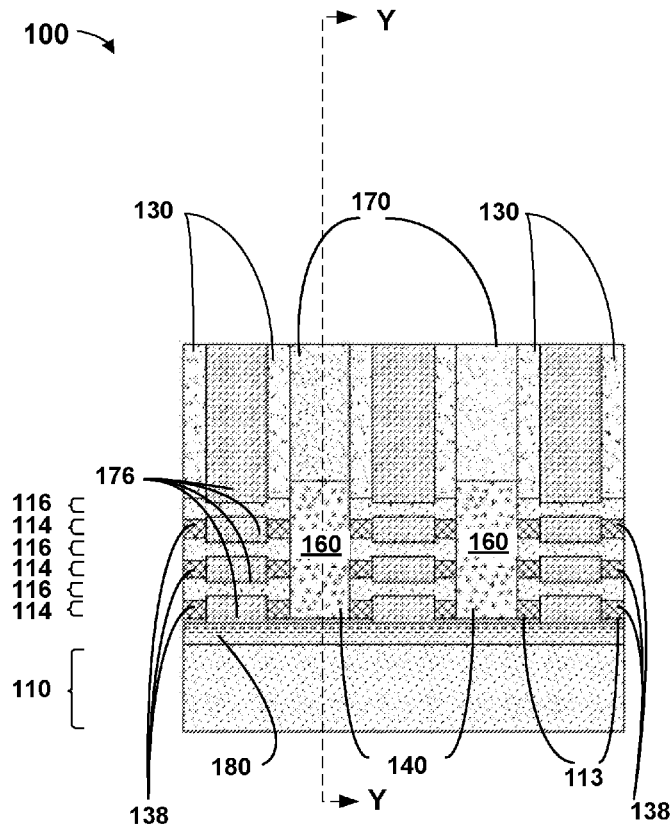
US 20230178617A1

(19) **United States**(12) **Patent Application Publication****Frougier et al.**(10) **Pub. No.: US 2023/0178617 A1**(43) **Pub. Date:****Jun. 8, 2023**(54) **NANOSHEET EPITAXY WITH FULL
BOTTOM ISOLATION**(71) Applicant: **INTERNATIONAL BUSINESS
MACHINES CORPORATION,**
ARMONK, NY (US)(72) Inventors: **Julien Frougier,** Albany, NY (US);
Andrew M. Greene, Slingerlands, NY
(US); **Ruilong Xie,** Niskayuna, NY
(US); **Lan Yu,** Voorheesville, NY (US);
PIETRO MONTANINI, Albany, NY
(US)(21) Appl. No.: **17/457,448**(22) Filed: **Dec. 3, 2021****Publication Classification**(51) **Int. Cl.**
H01L 29/423 (2006.01)
H01L 29/786 (2006.01)
H01L 27/092 (2006.01)
H01L 29/06 (2006.01)
H01L 29/417 (2006.01)
H01L 21/8238 (2006.01)(52) **U.S. Cl.**CPC .. **H01L 29/42392** (2013.01); **H01L 29/78696**
(2013.01); **H01L 27/092** (2013.01); **H01L**
29/0665 (2013.01); **H01L 29/41775** (2013.01);
H01L 21/823807 (2013.01); **H01L 21/823878**
(2013.01)

(57)

ABSTRACT

Semiconductor channel layers vertically aligned and stacked one on top of another, separated by a gate stack material wrapping around the semiconductor channel layers, a heavily doped p-type field effect transistor (p-FET) source drain epitaxy region adjacent to the semiconductor channel layers, a horizontal lower surface of the p-FET source drain epitaxy region is adjacent to a horizontal upper surface of an undoped silicon epitaxy. Forming a first stack, second stack and third stack of nanosheet layers on a substrate, each including alternating layers of a sacrificial and a semiconductor channel vertically aligned and stacked one on top of another, forming a first sacrificial gate across the first stack, a second sacrificial gate across the second stack and a third sacrificial gate across the third stack, forming an undoped silicon epitaxy between the first and the second stacks and between the second and the third stacks.

**Section X1-X1**

100 ↗

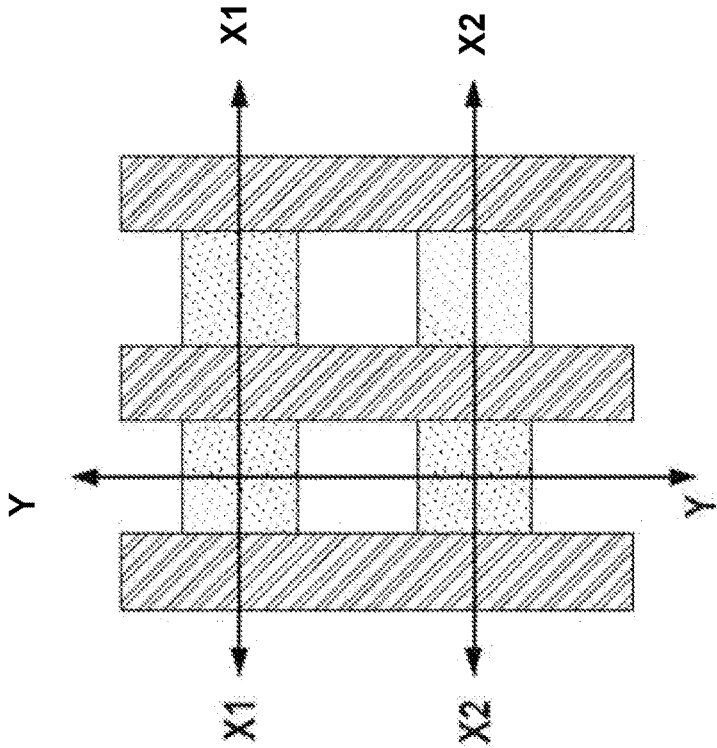


Figure 1

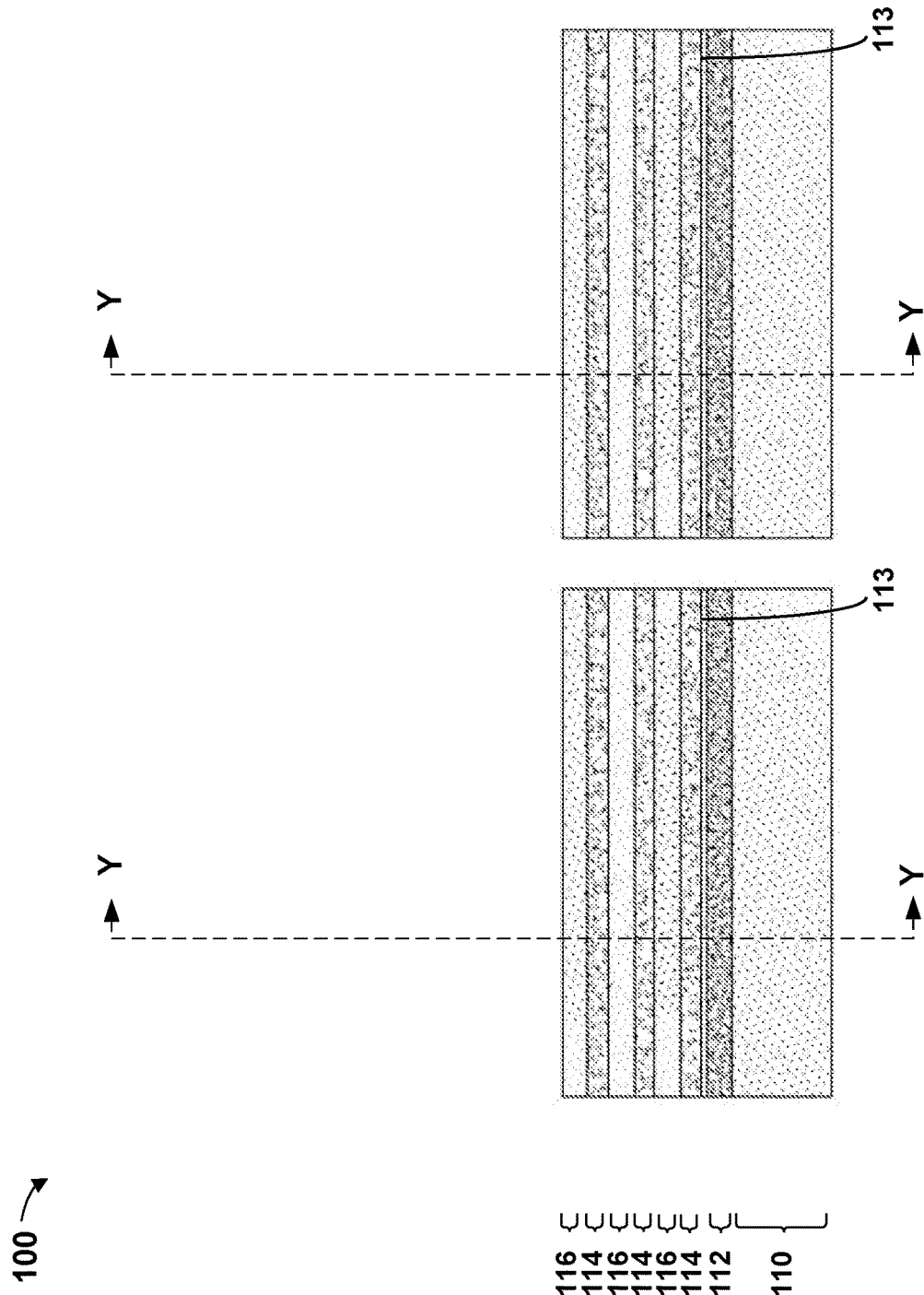


Figure 3
Section X2-X2

Figure 2
Section X1-X1

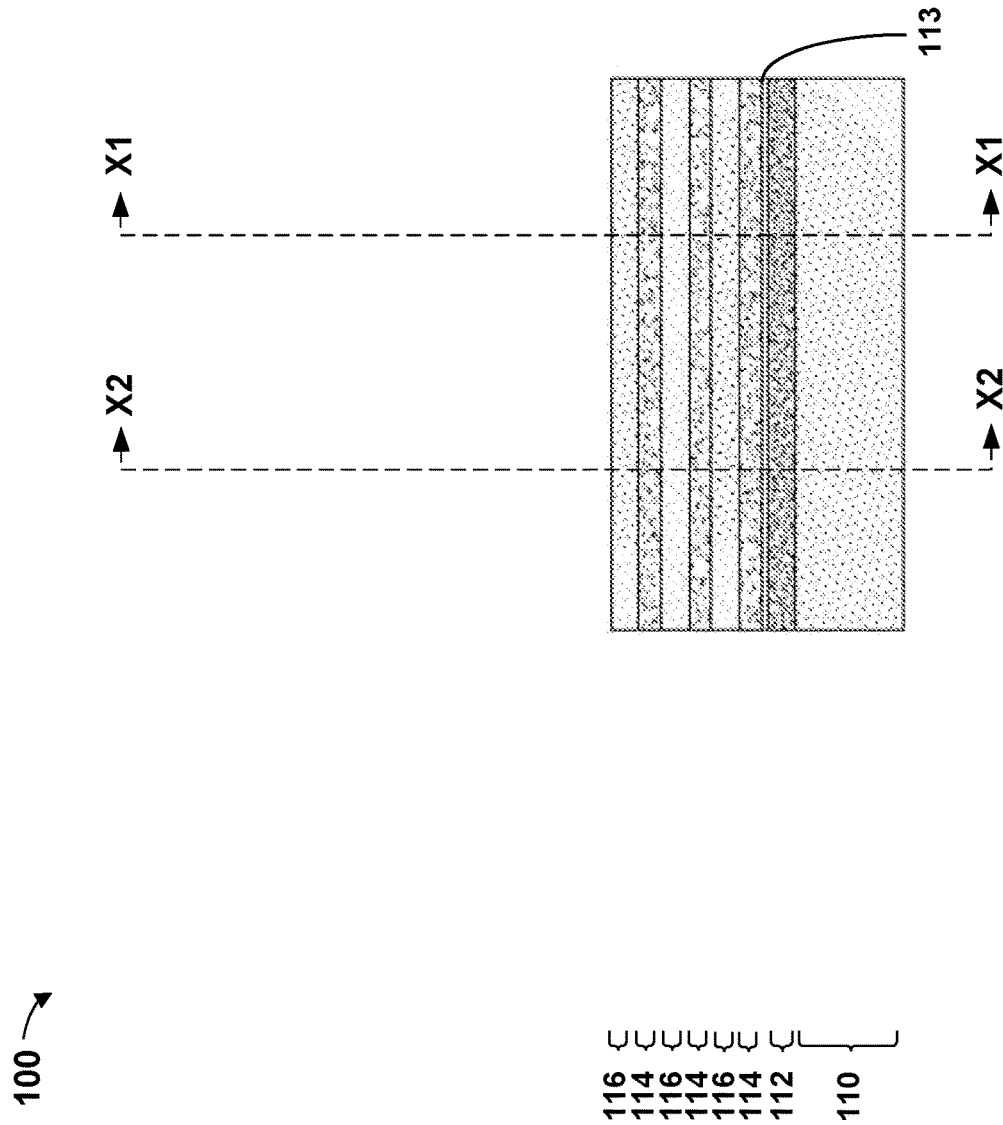


Figure 4
Section Y-Y

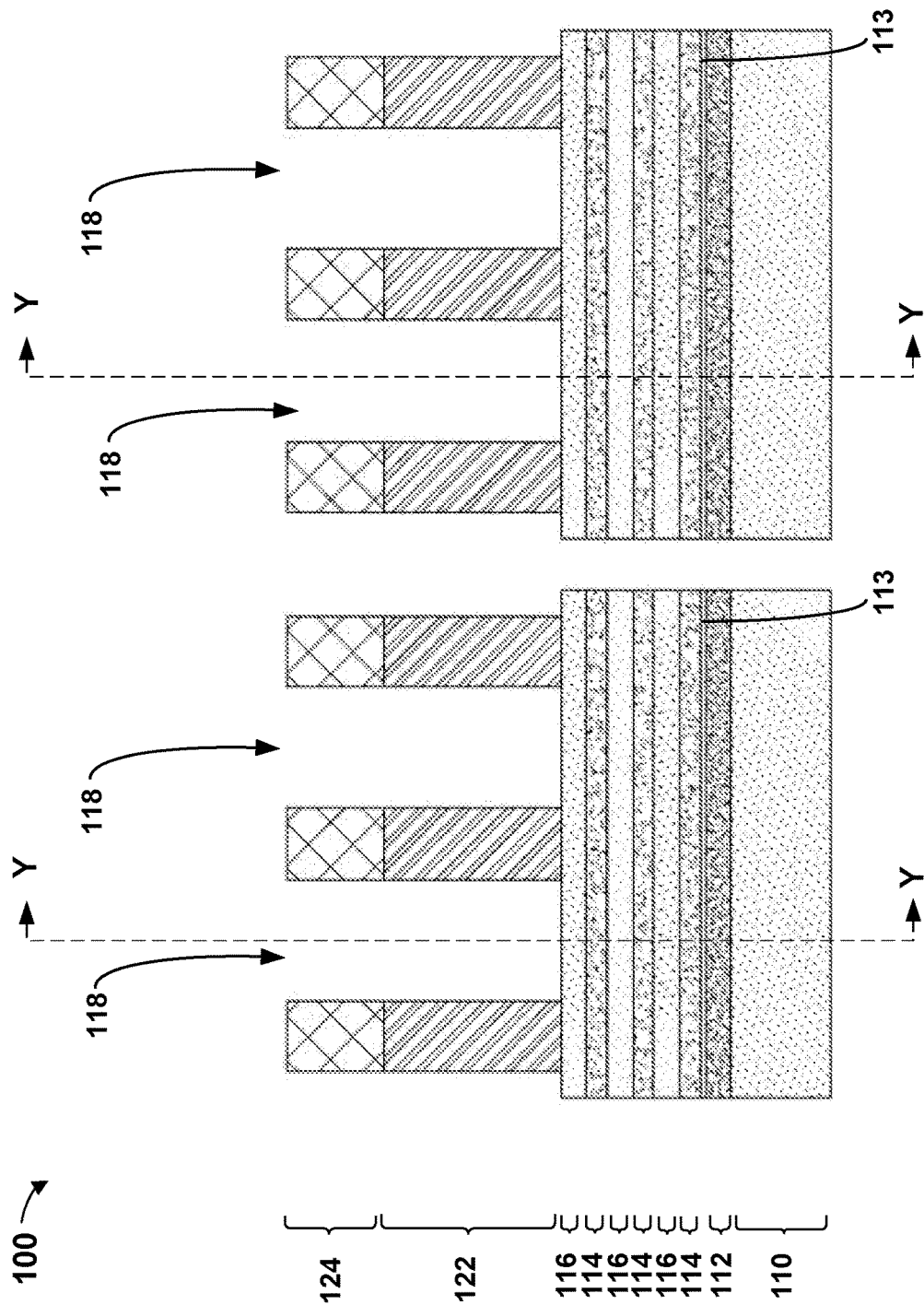


Figure 5
Section X1-X1

Figure 6
Section X2-X2

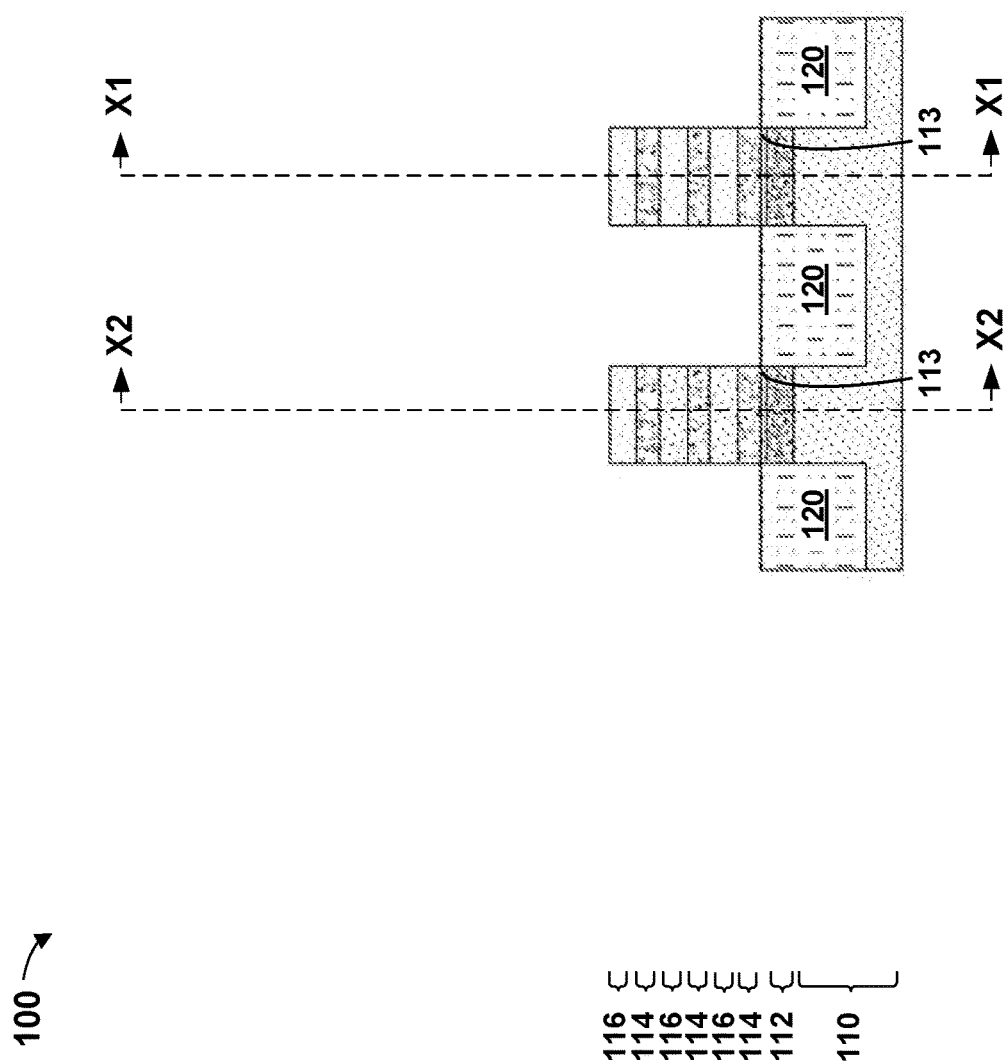


Figure 7
Section Y-Y

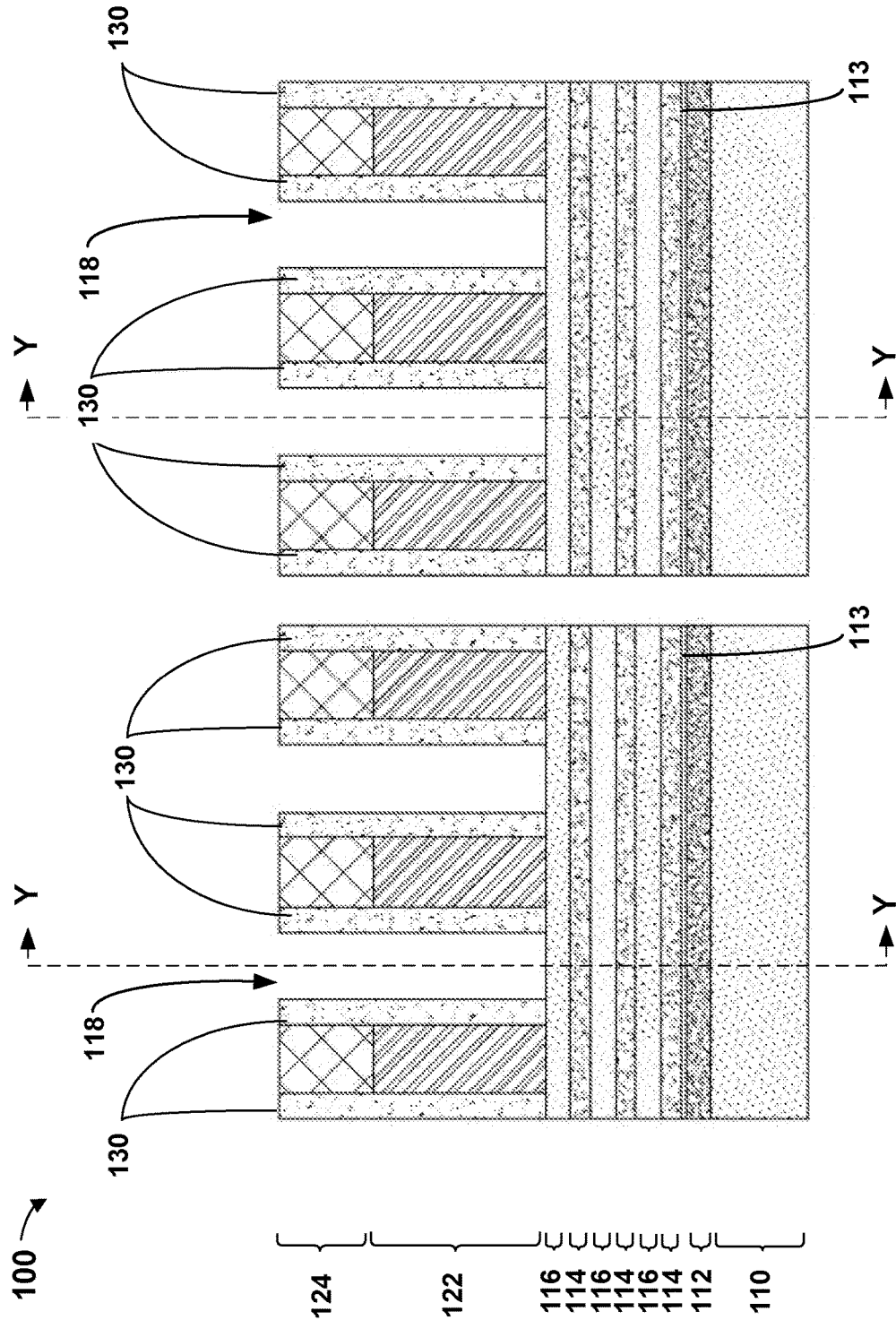


Figure 8
Section X1-X1

Figure 9
Section X2-X2

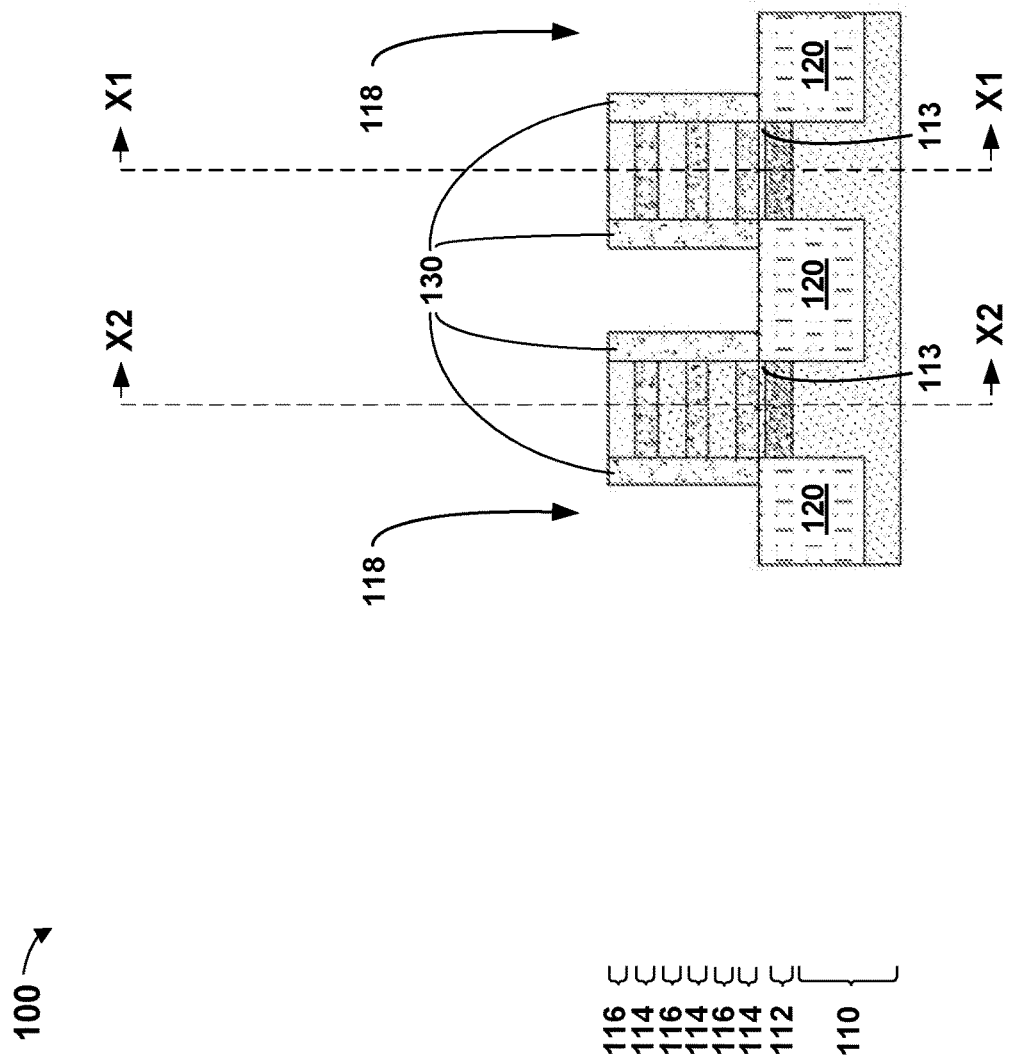


Figure 10
Section Y-Y

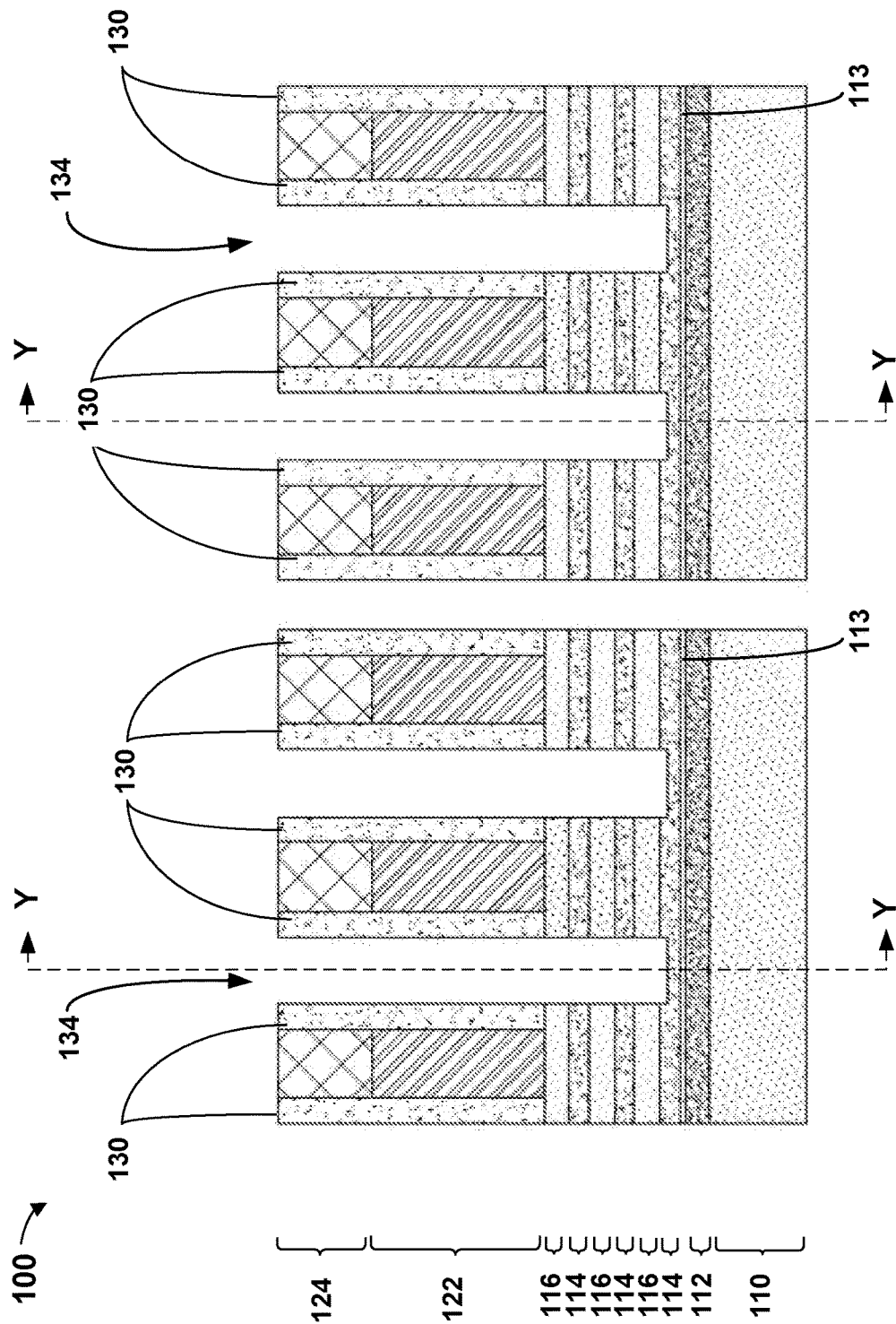


Figure 12
Section X2-X2

Figure 11
Section X1-X1

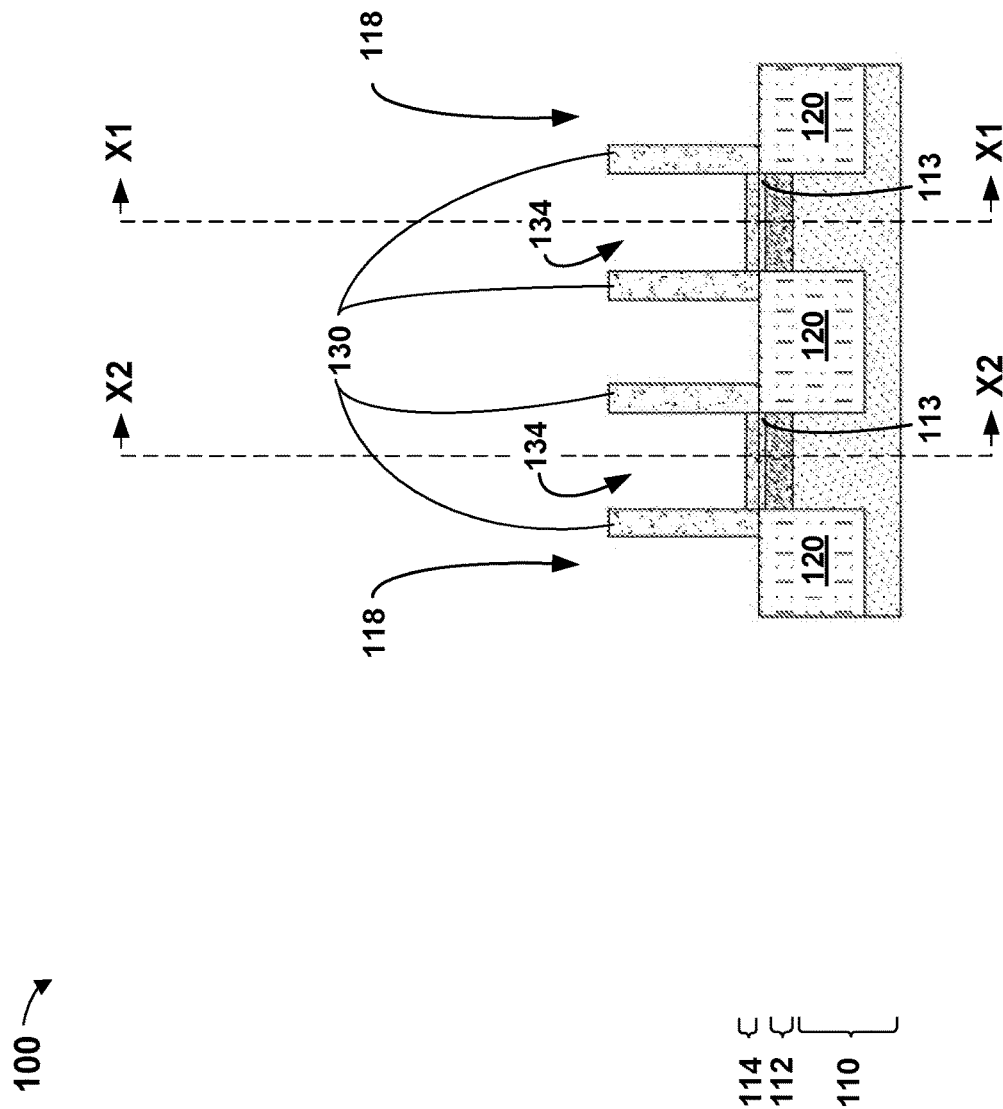


Figure 13
Section Y-Y

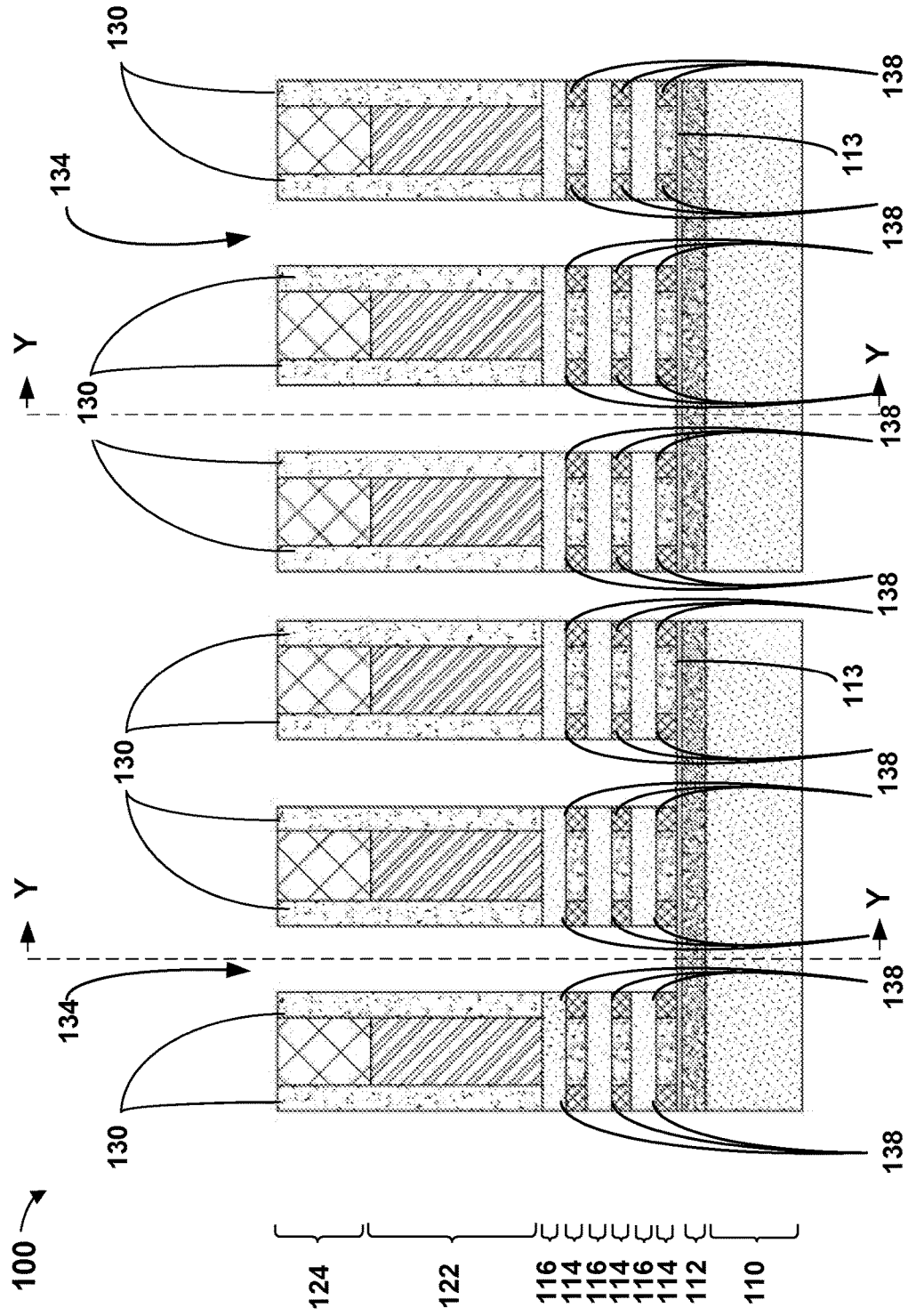


Figure 15
Section X2-X2

Figure 14
Section X1-X1

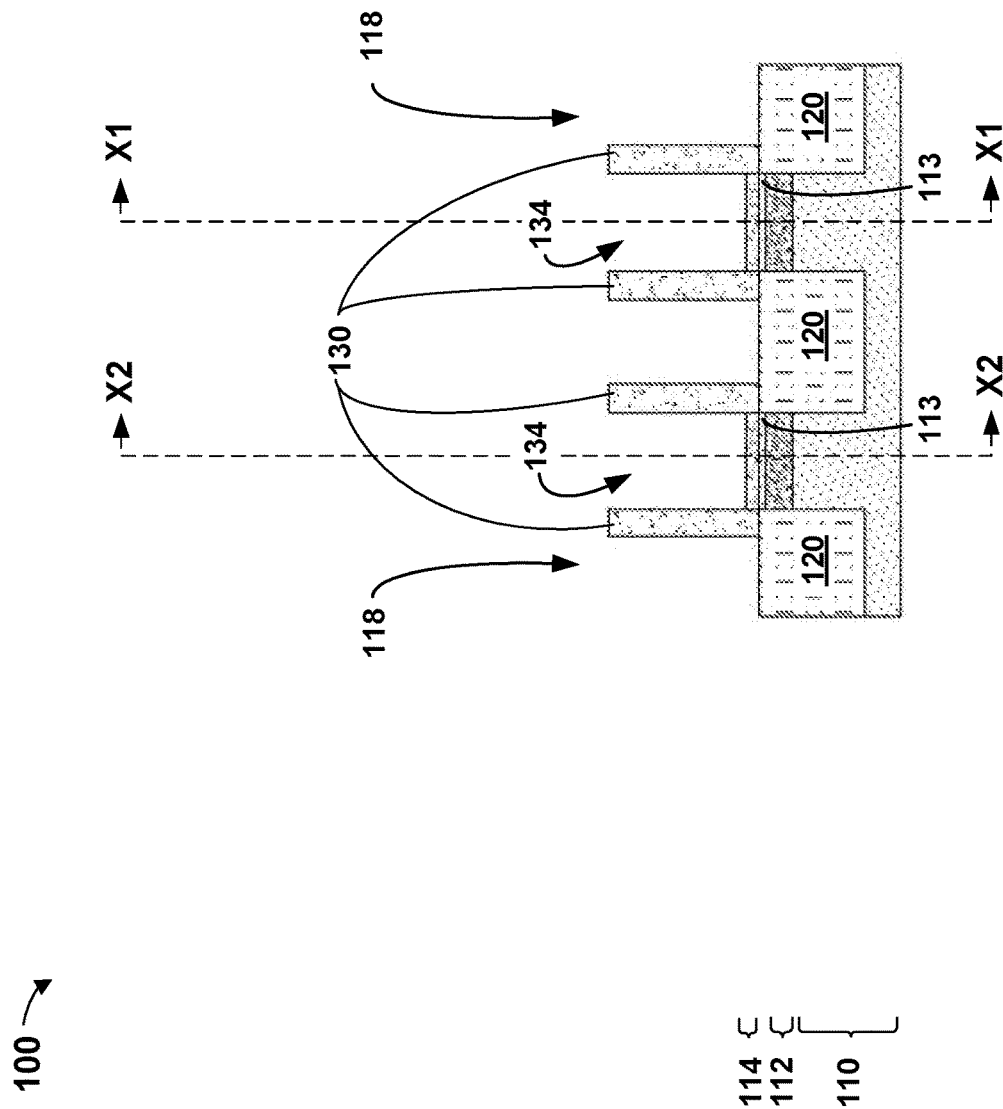


Figure 16
Section Y-Y

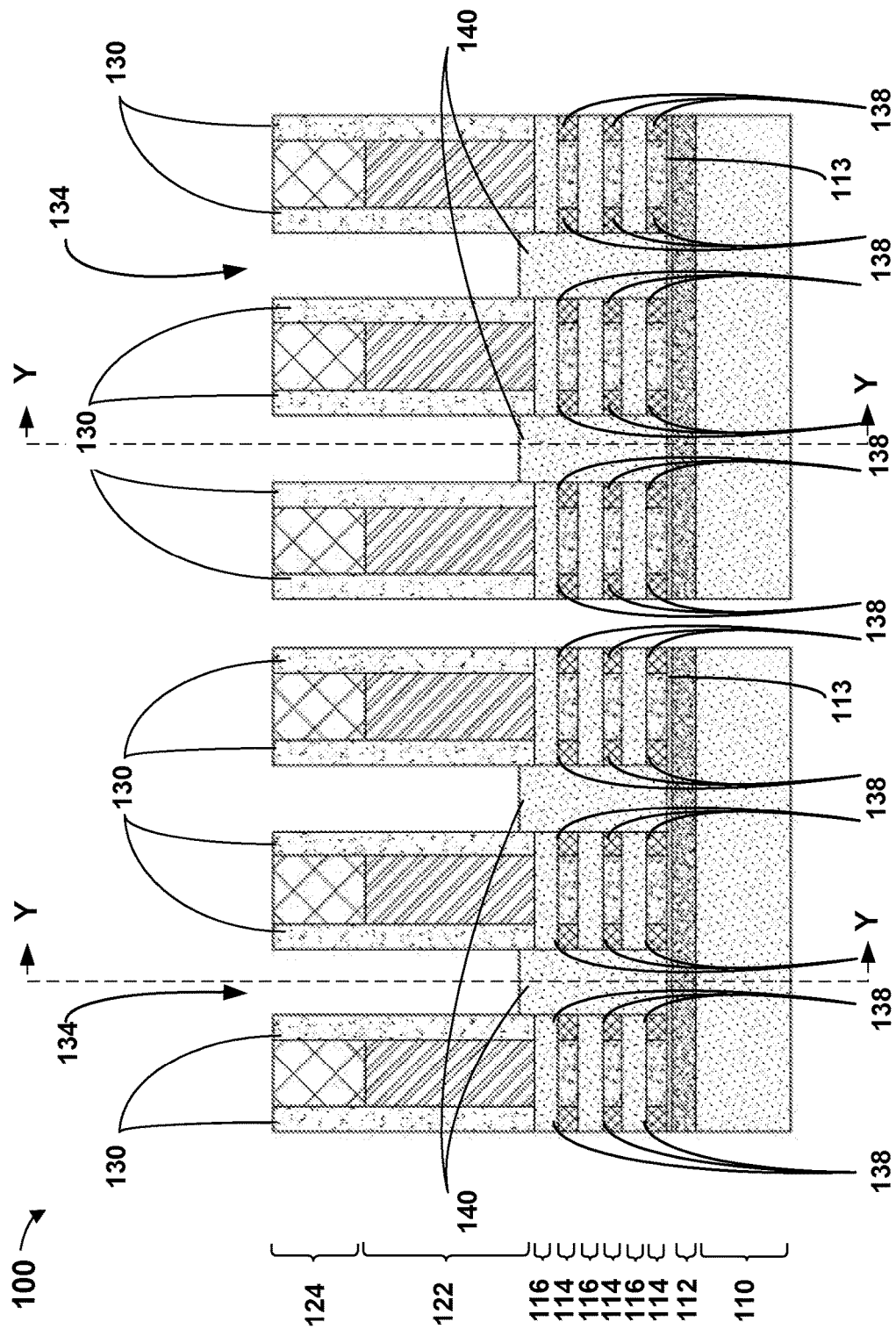
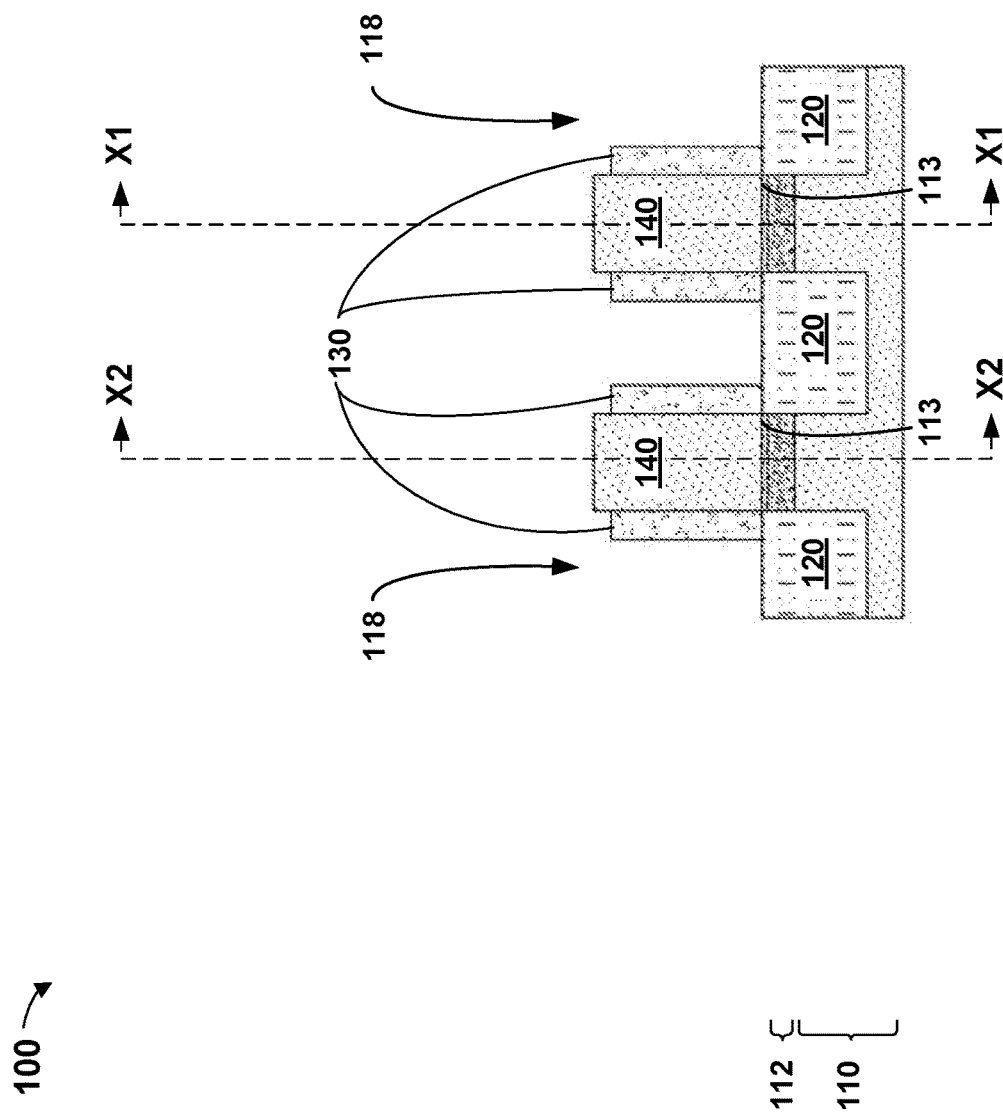


Figure 18
Section X2-X2

Figure 17
Section X1-X1



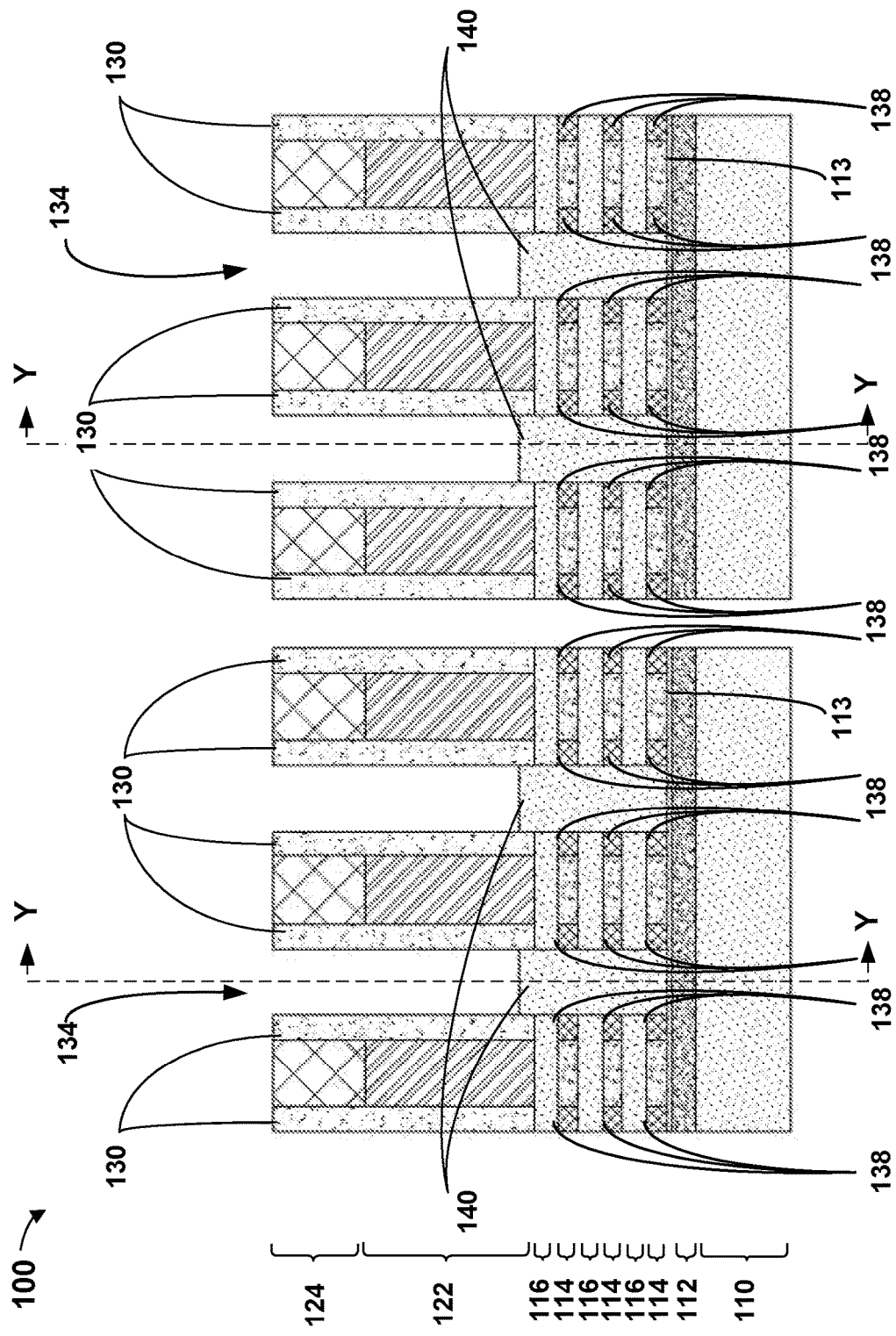


Figure 21
Section X2-X2

Figure 20
Section X1-X1

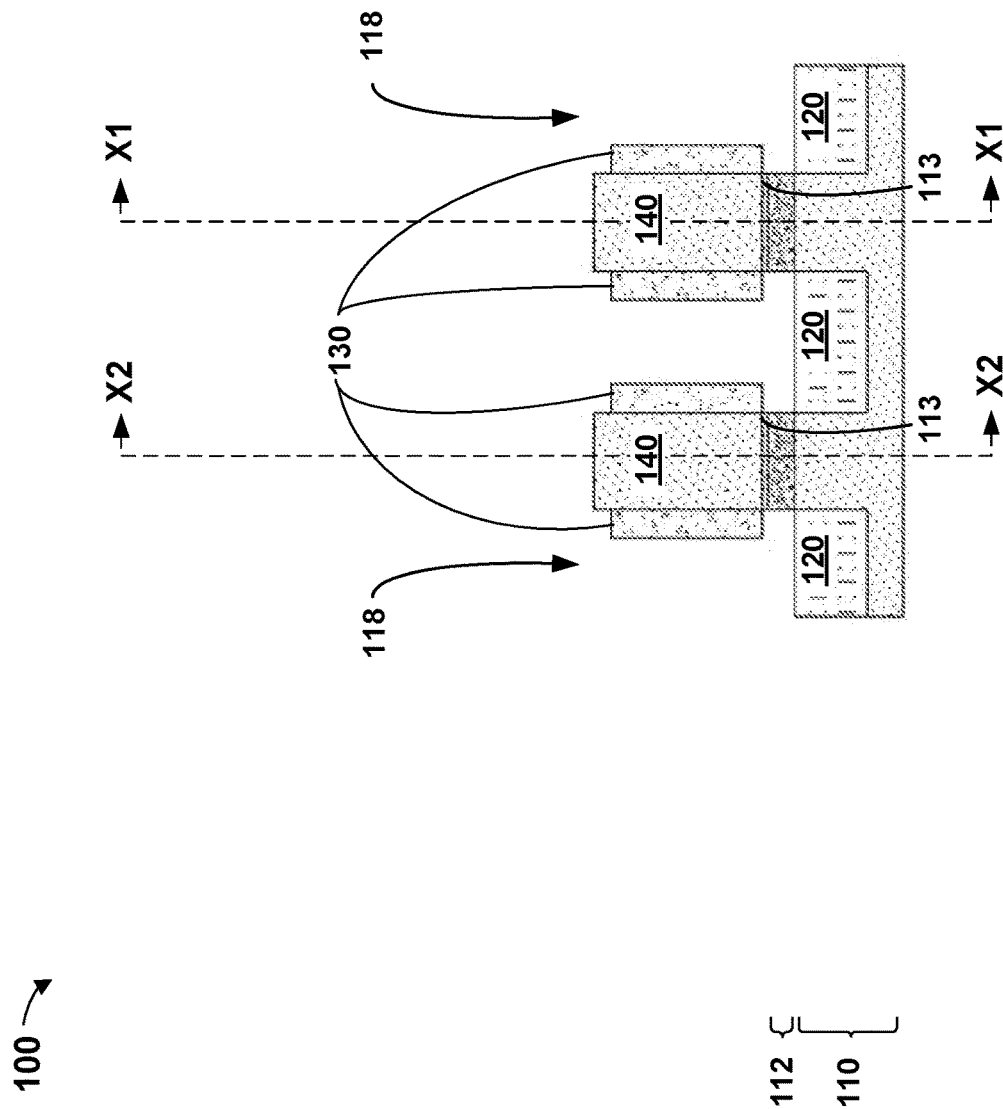


Figure 22
Section Y-Y

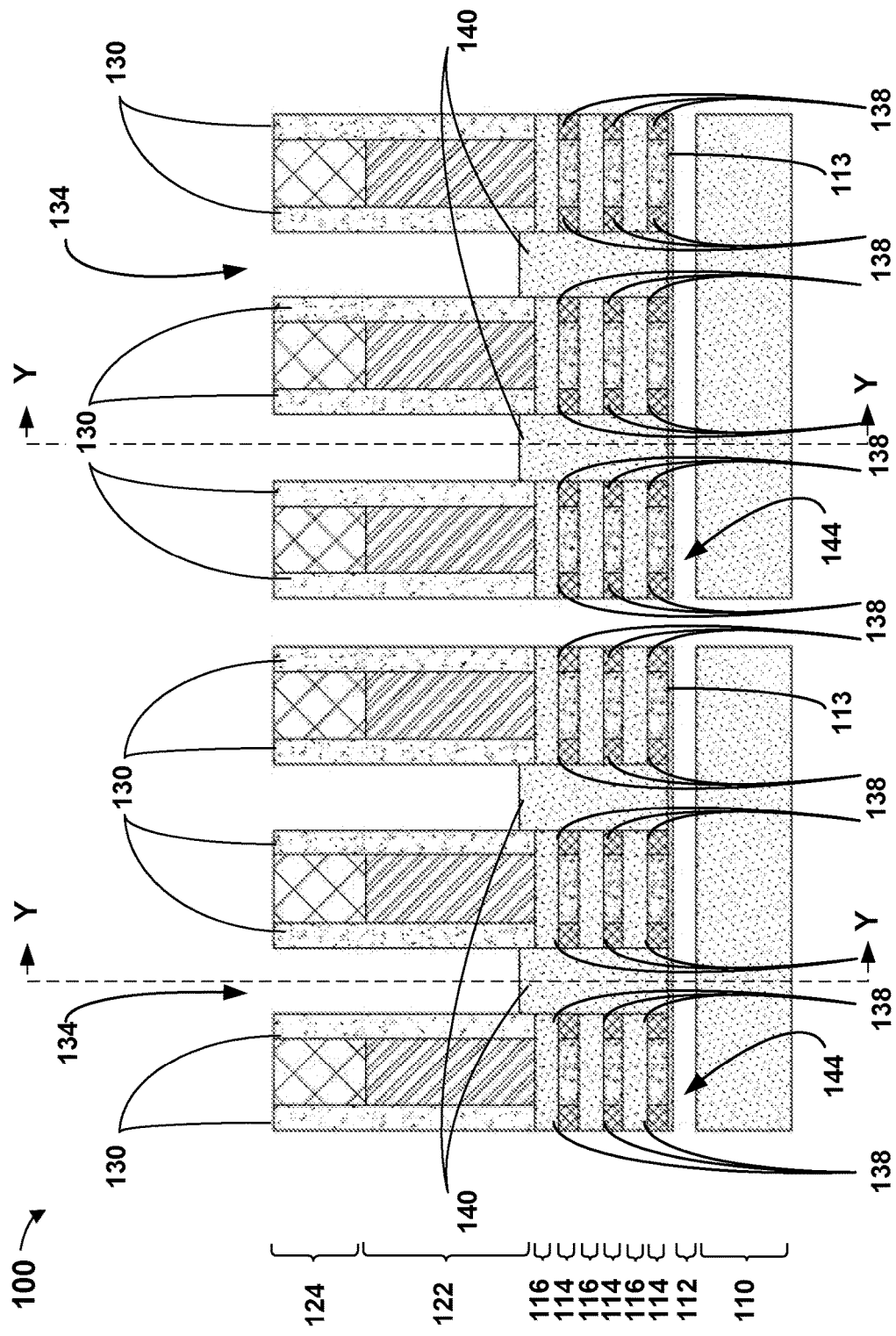


Figure 23
Section X1-X1

Figure 24
Section X2-X2

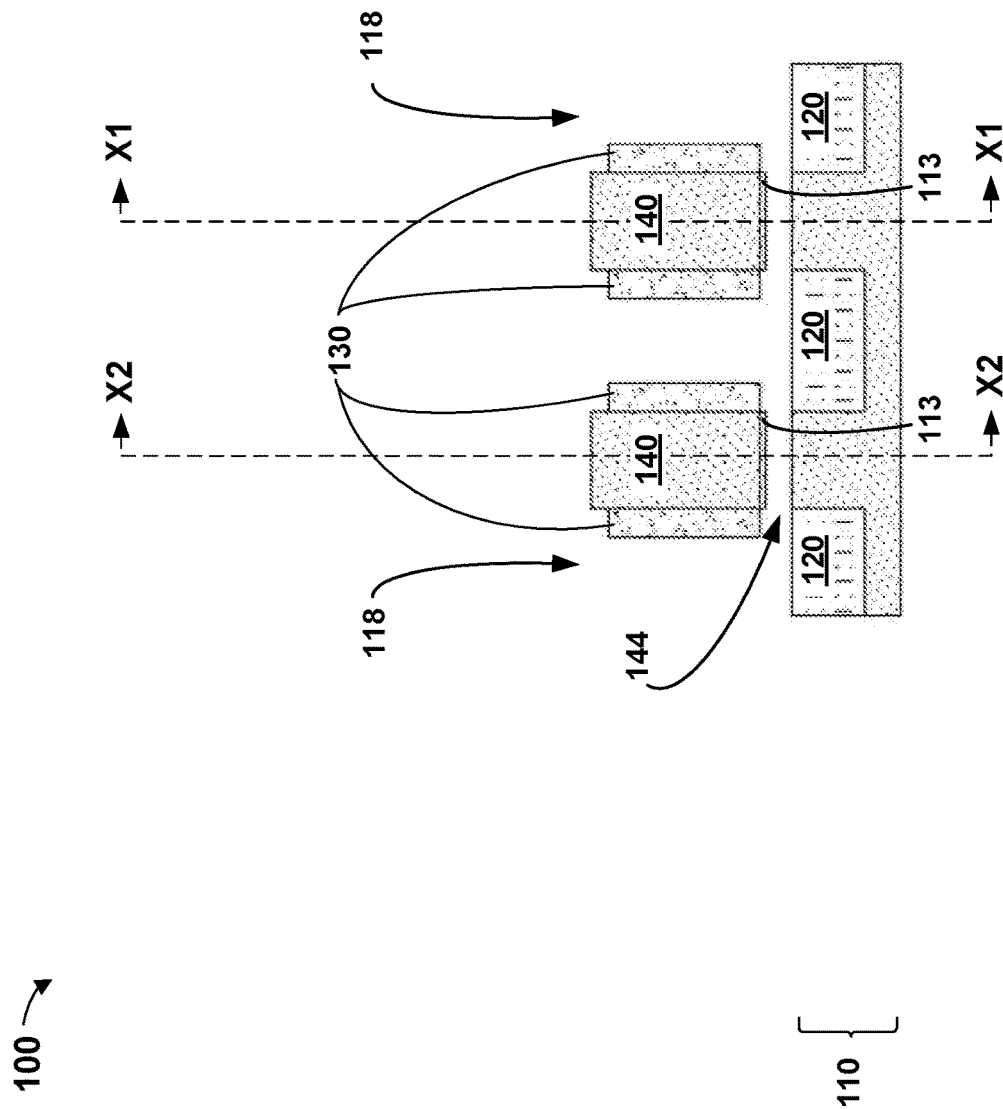


Figure 25
Section Y-Y

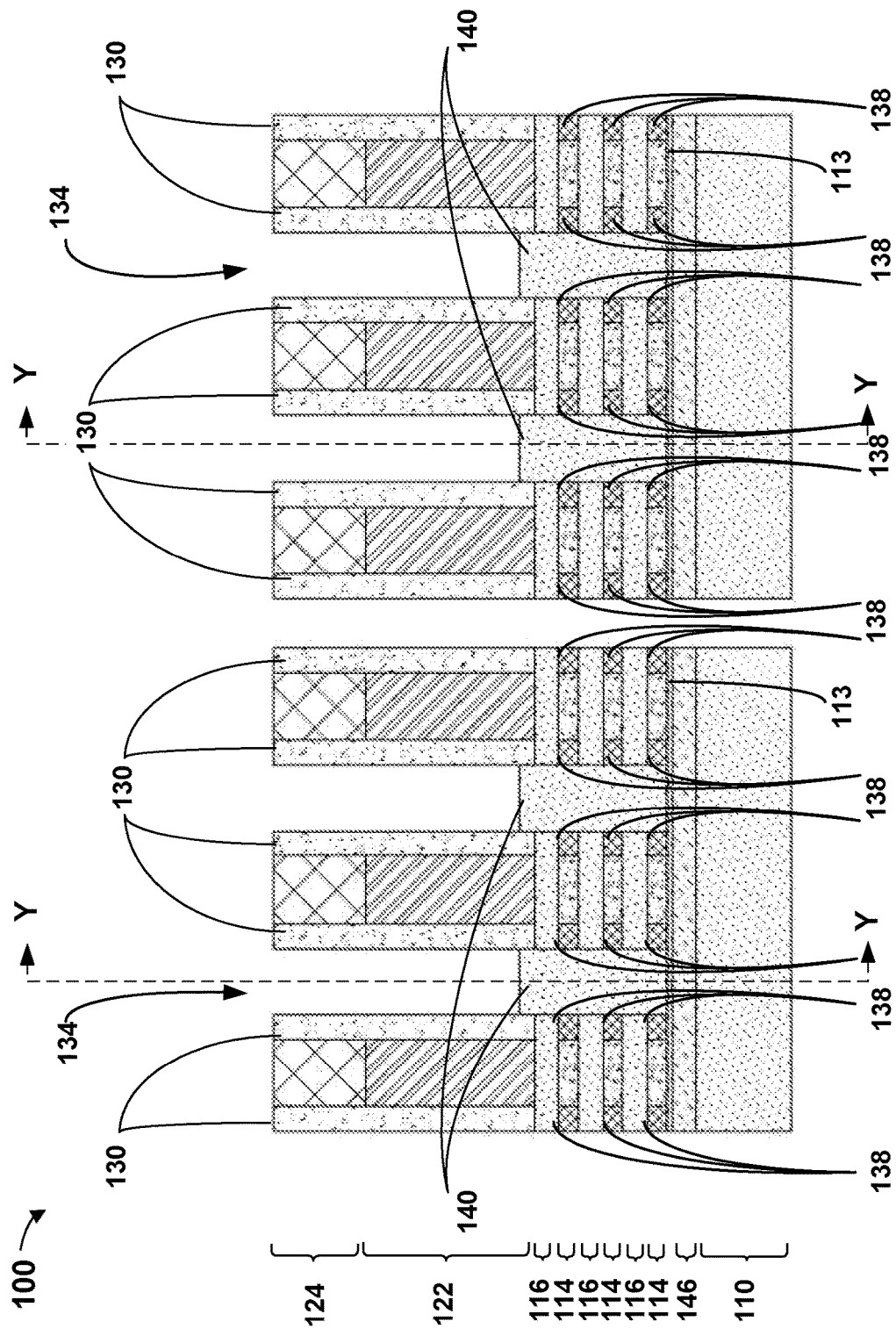


Figure 26
Section X1-X1

Figure 27
Section X2-X2

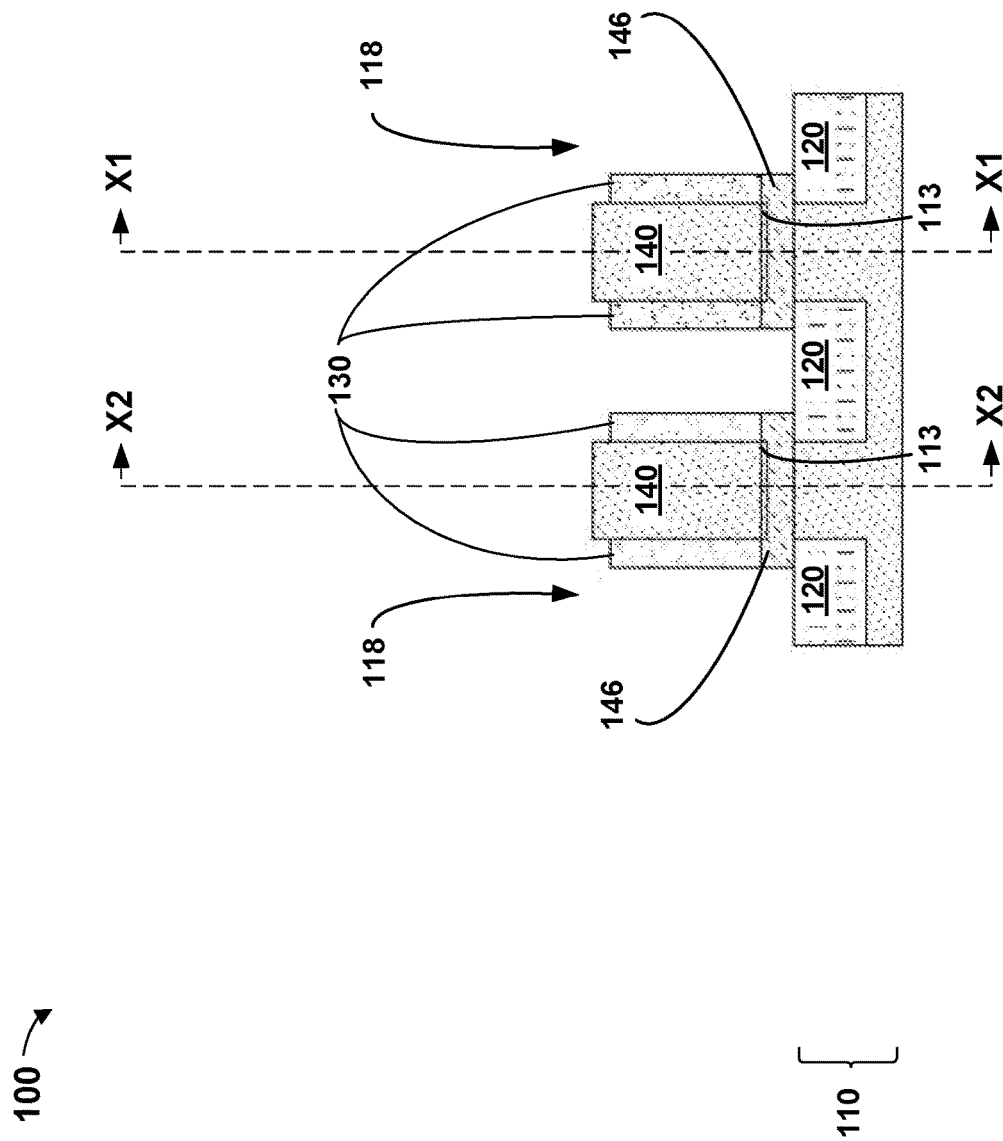


Figure 28
Section Y-Y

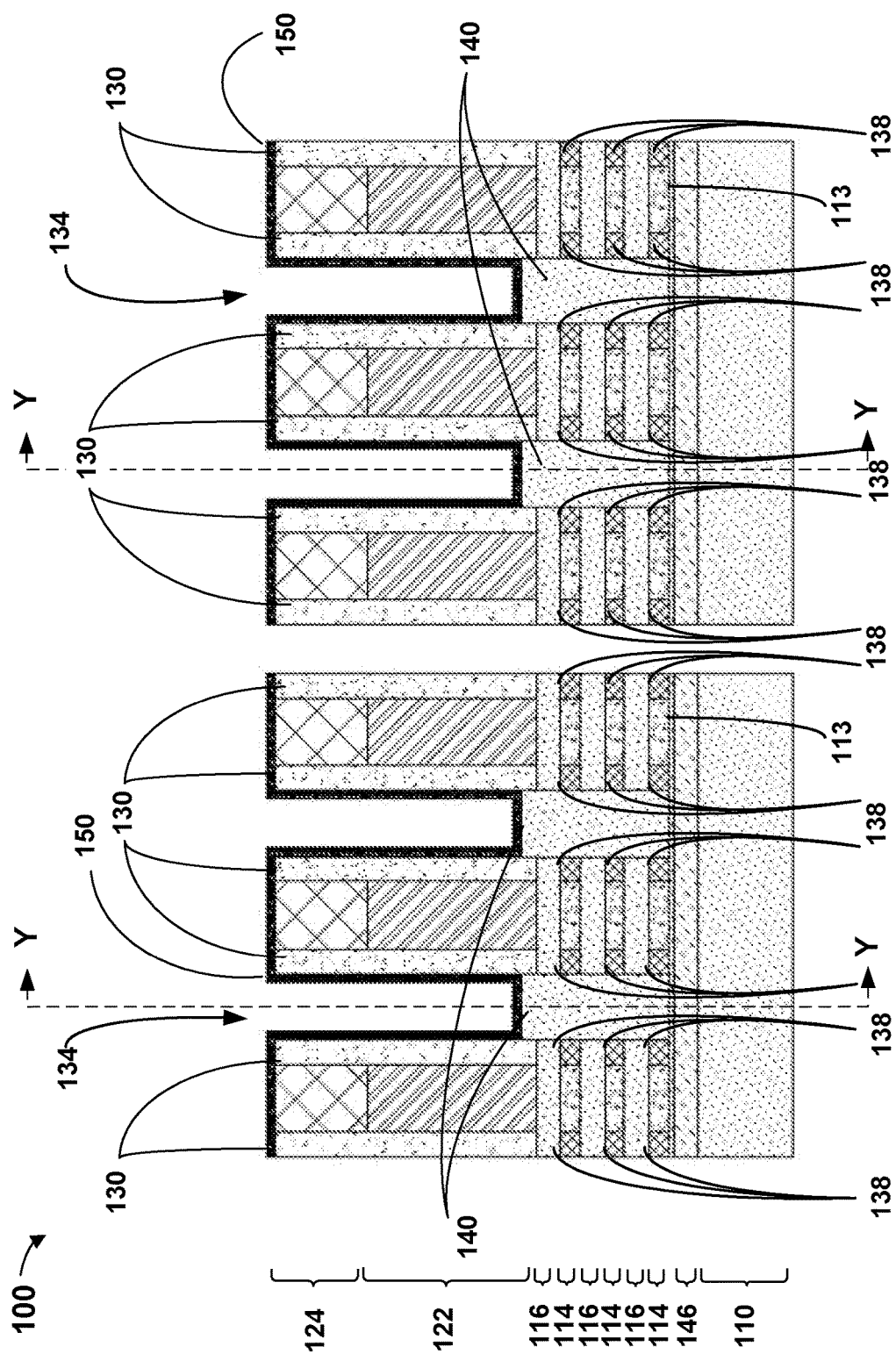


Figure 29
Section X1-X1

Figure 30
Section X2-X2

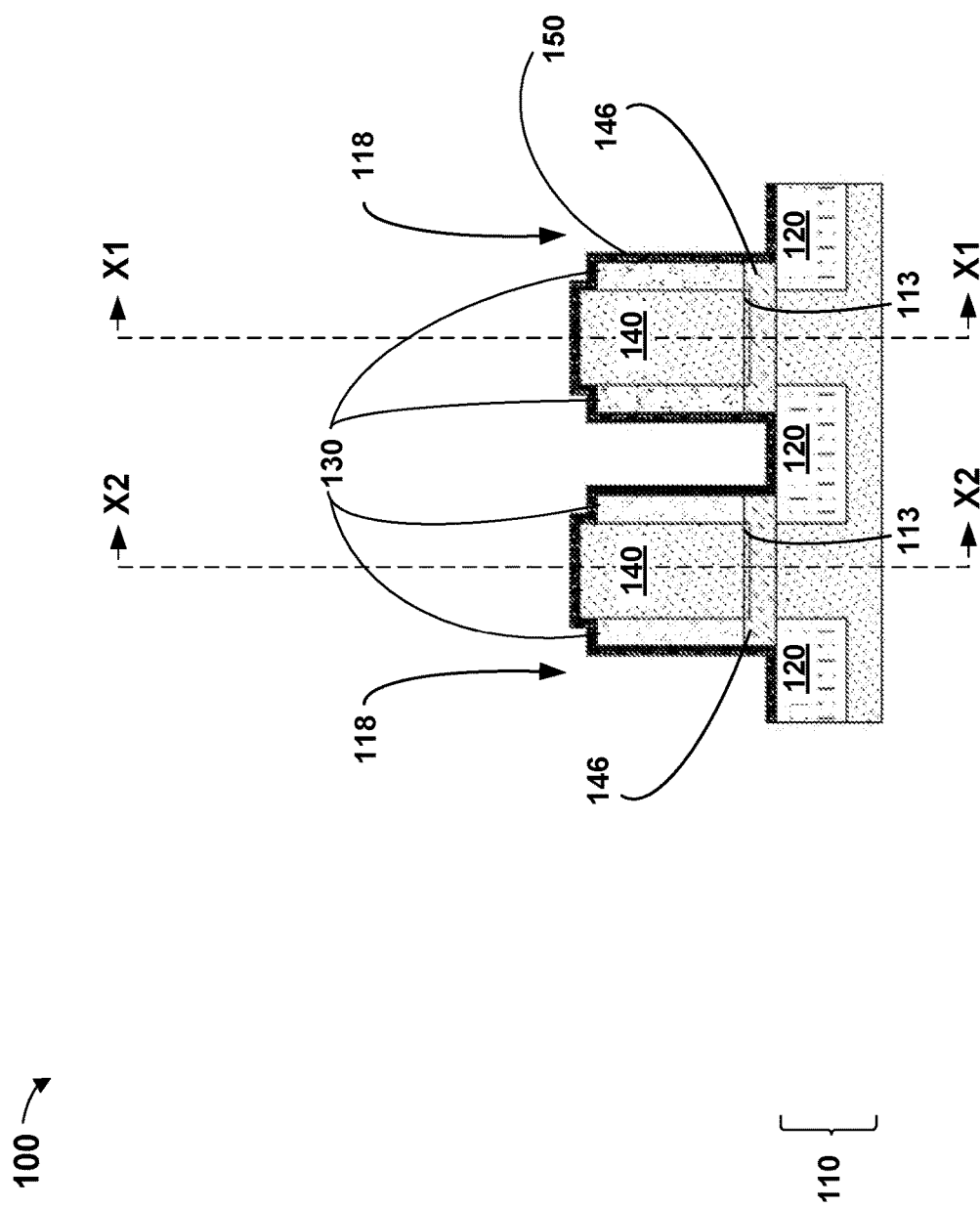


Figure 31
Section Y-Y

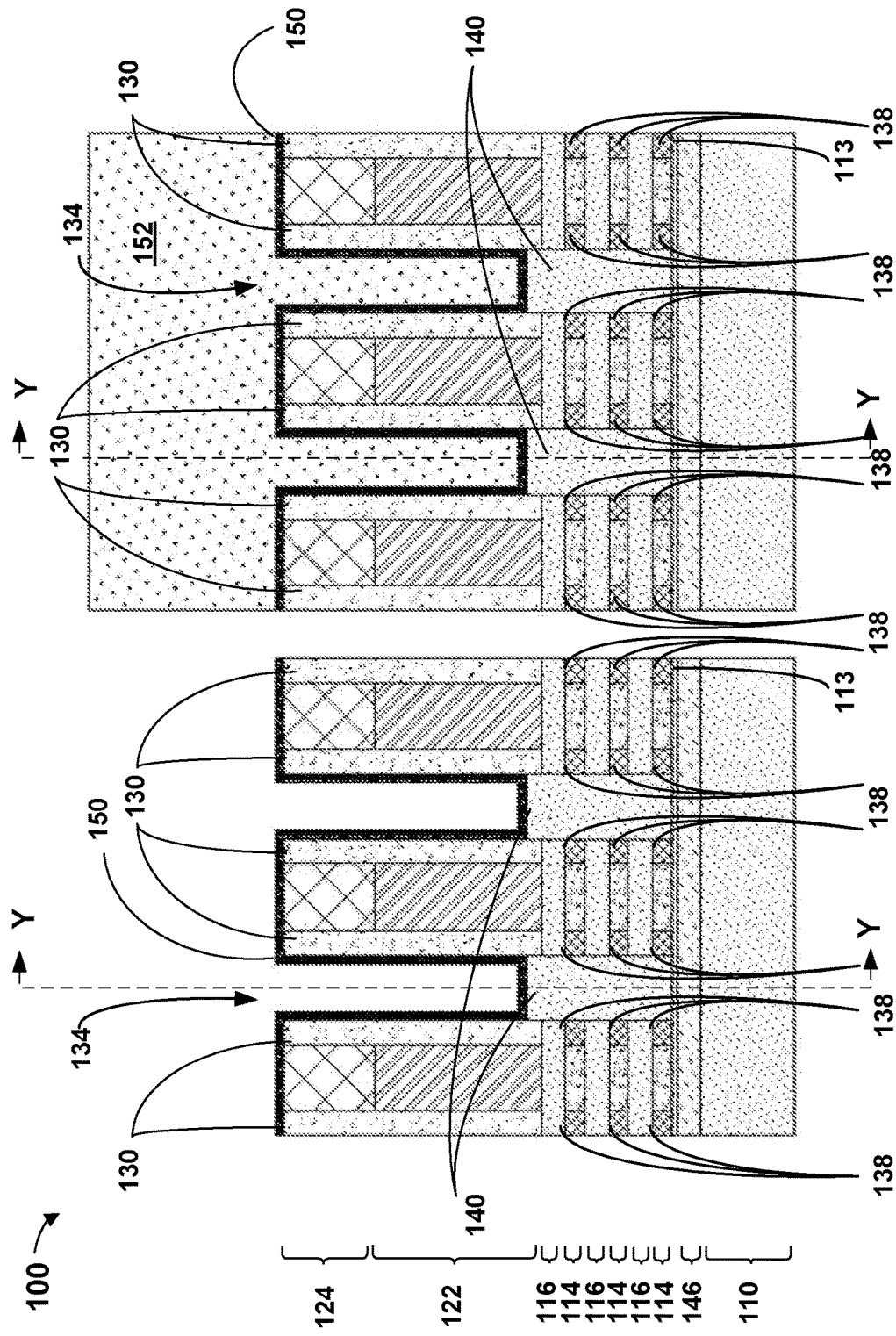


Figure 32
Section X1-X1

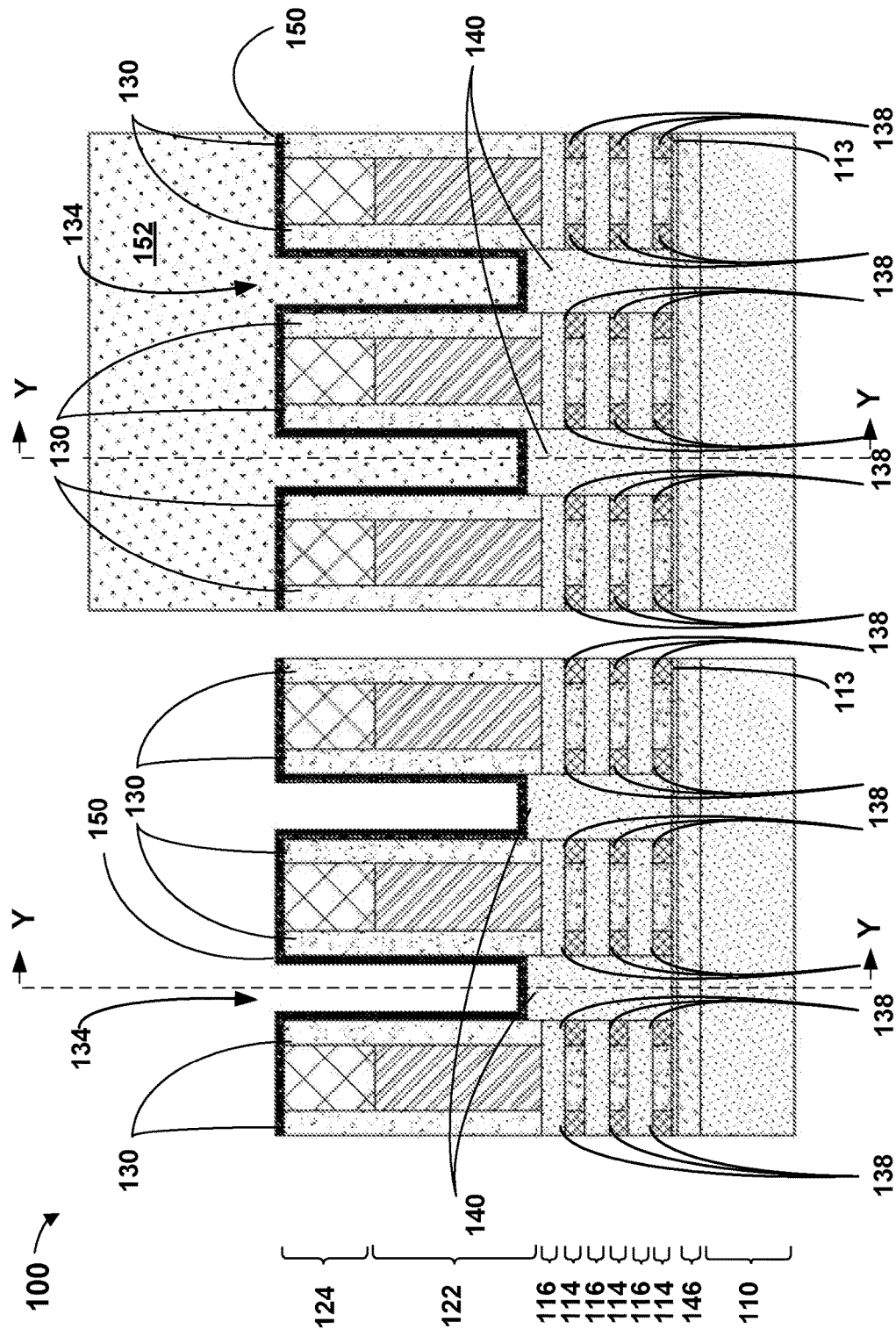


Figure 33
Section X2-X2

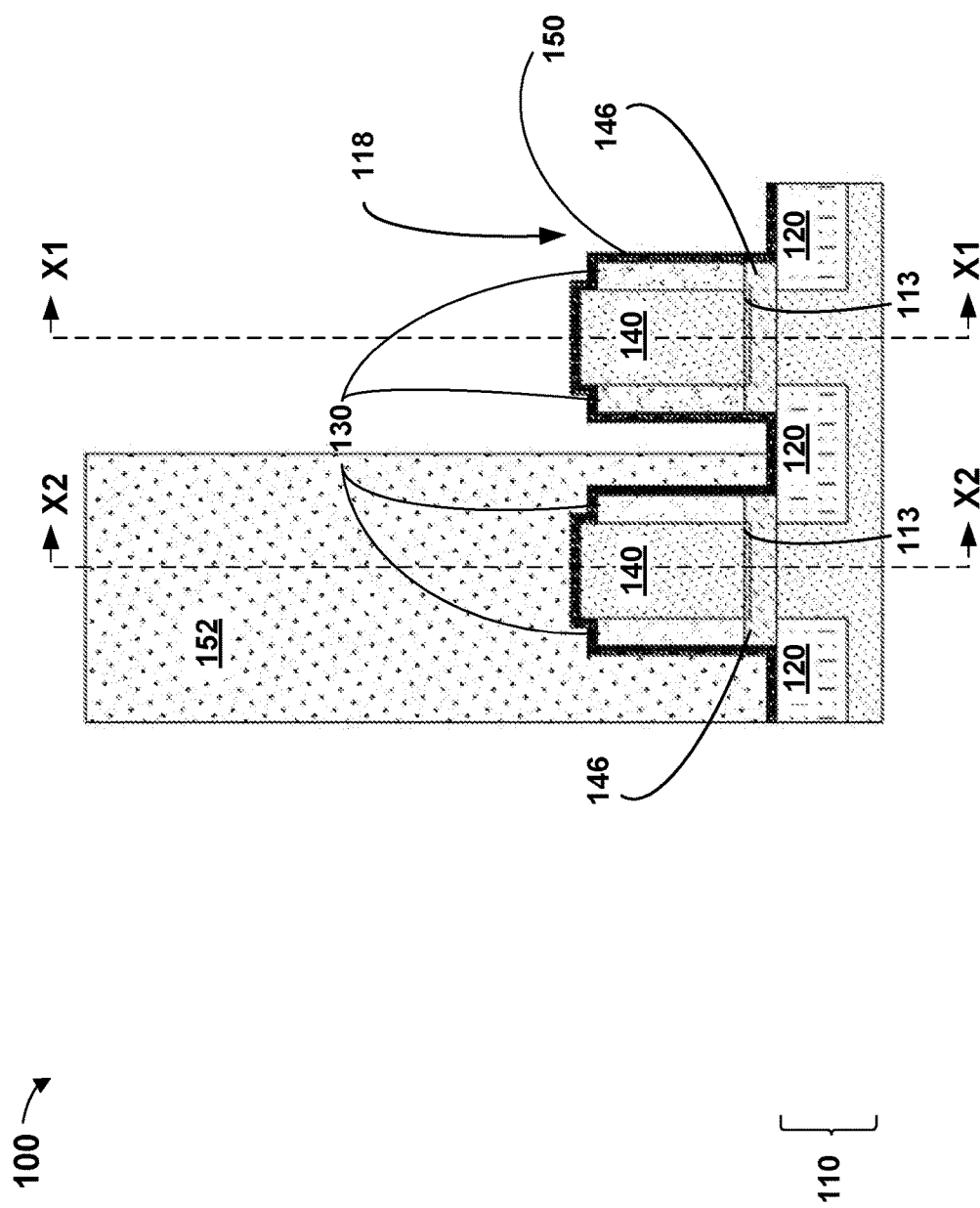


Figure 34
Section Y-Y

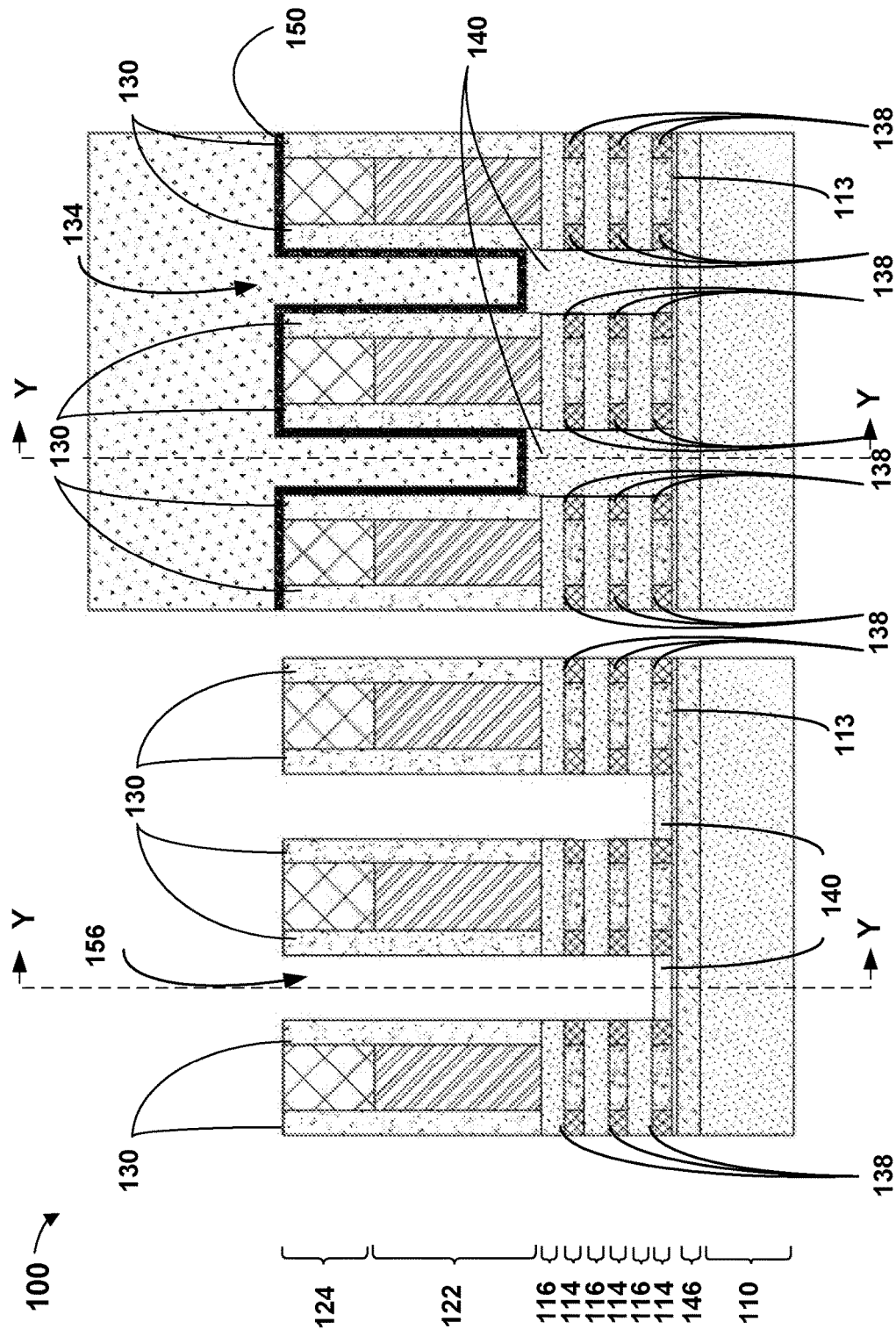


Figure 35
Section X1-X1

Figure 36
Section X2-X2

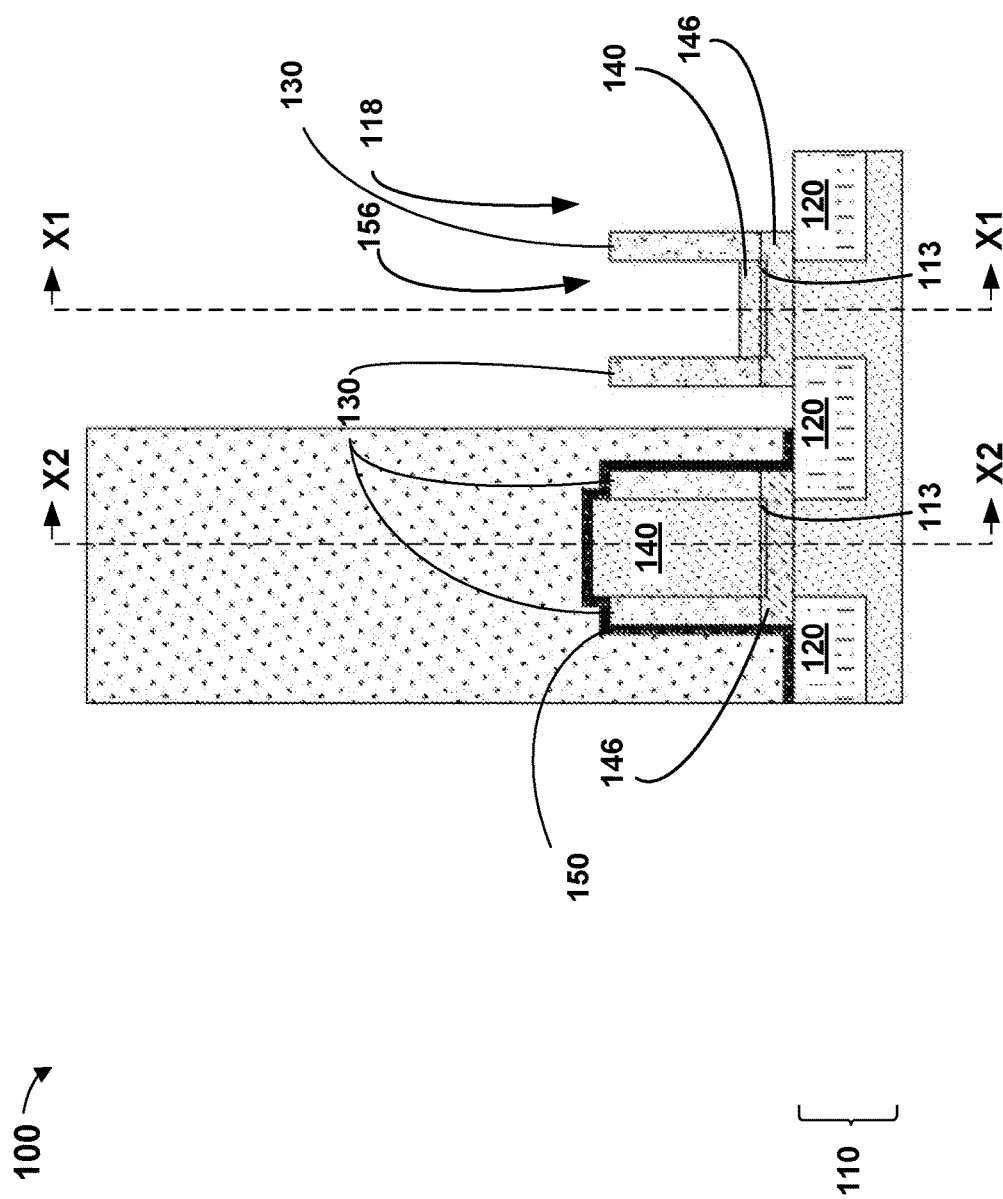


Figure 37

Section Y-Y

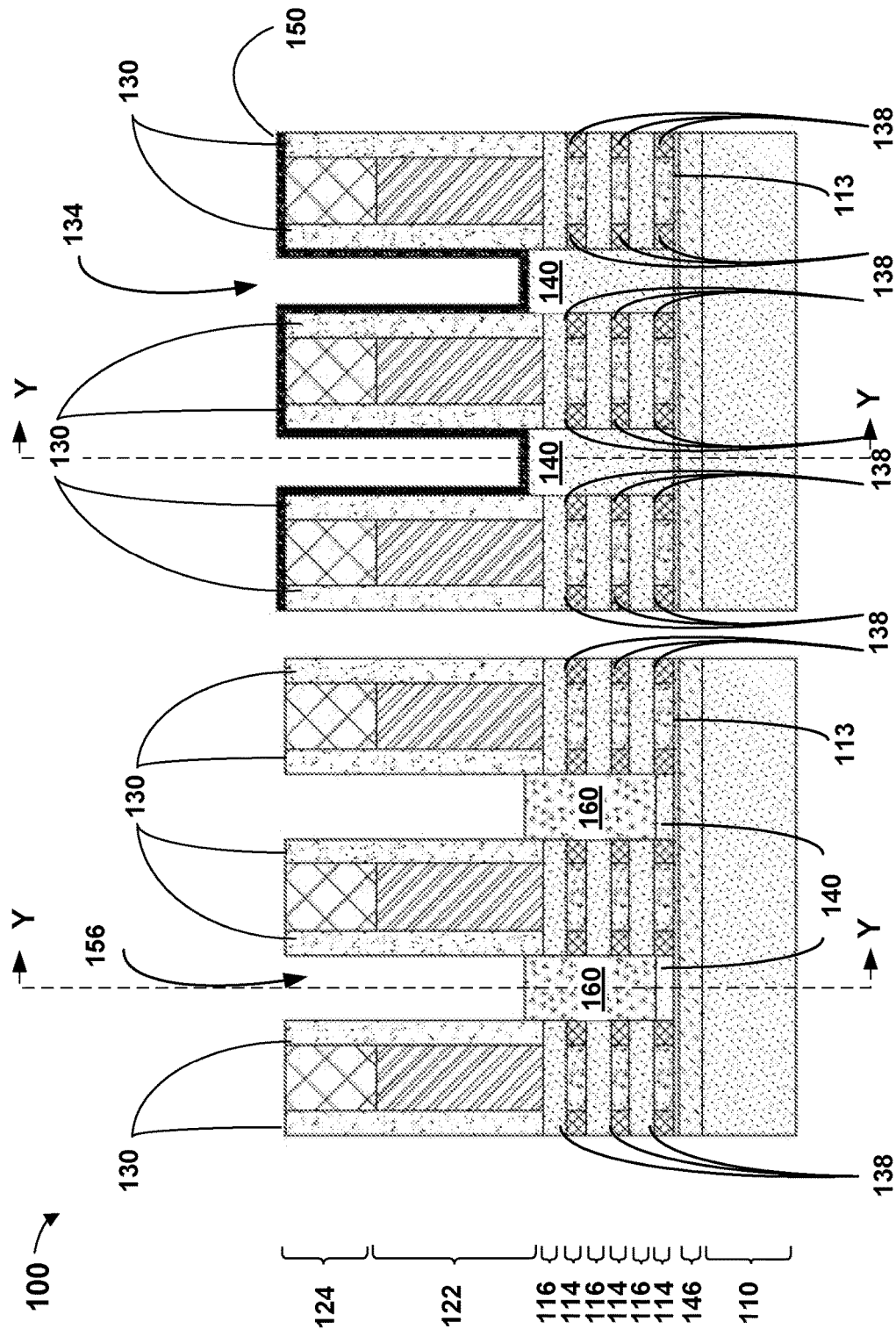


Figure 39
Section X2-X2

Figure 38
Section X1-X1

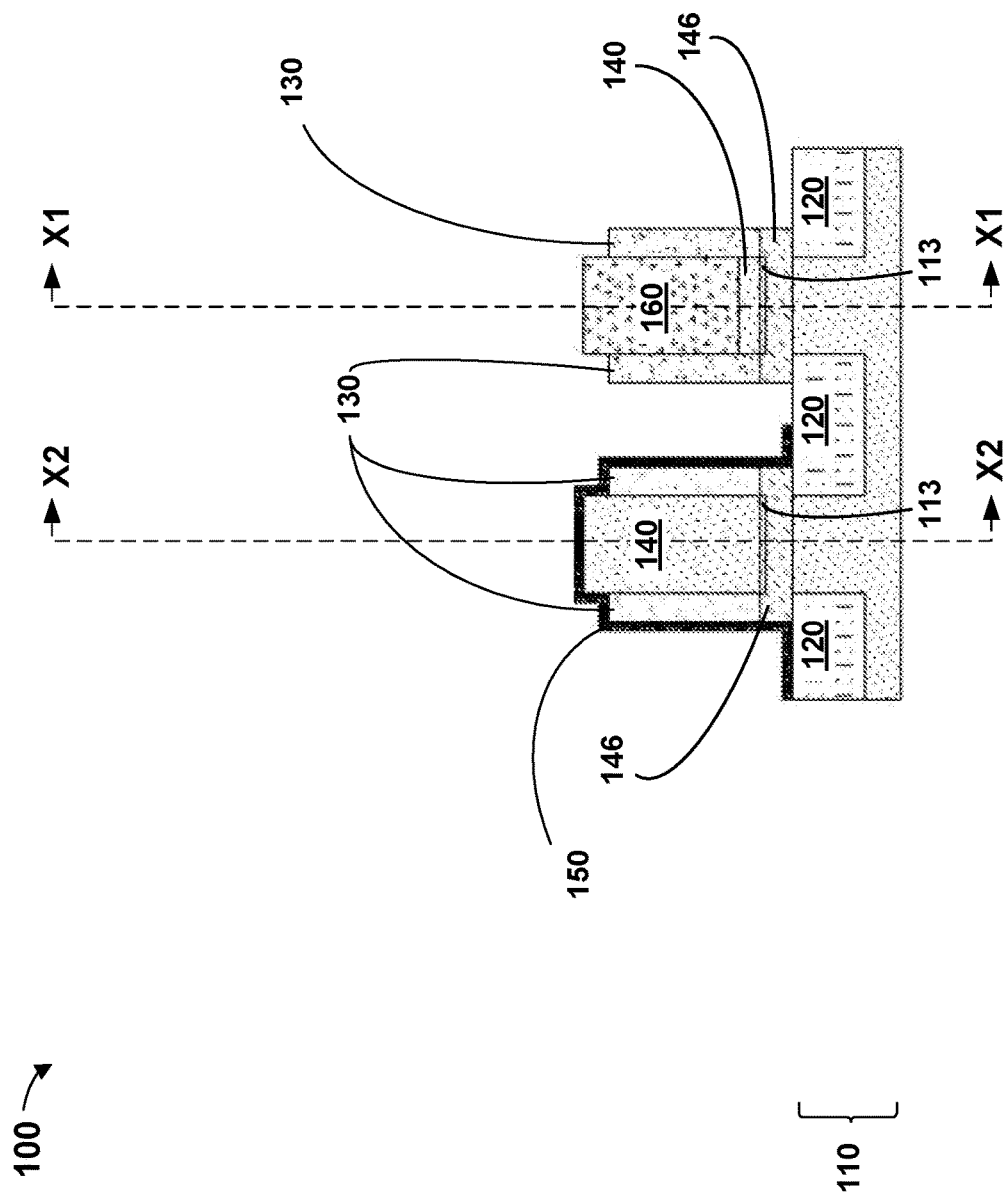


Figure 40
Section Y-Y

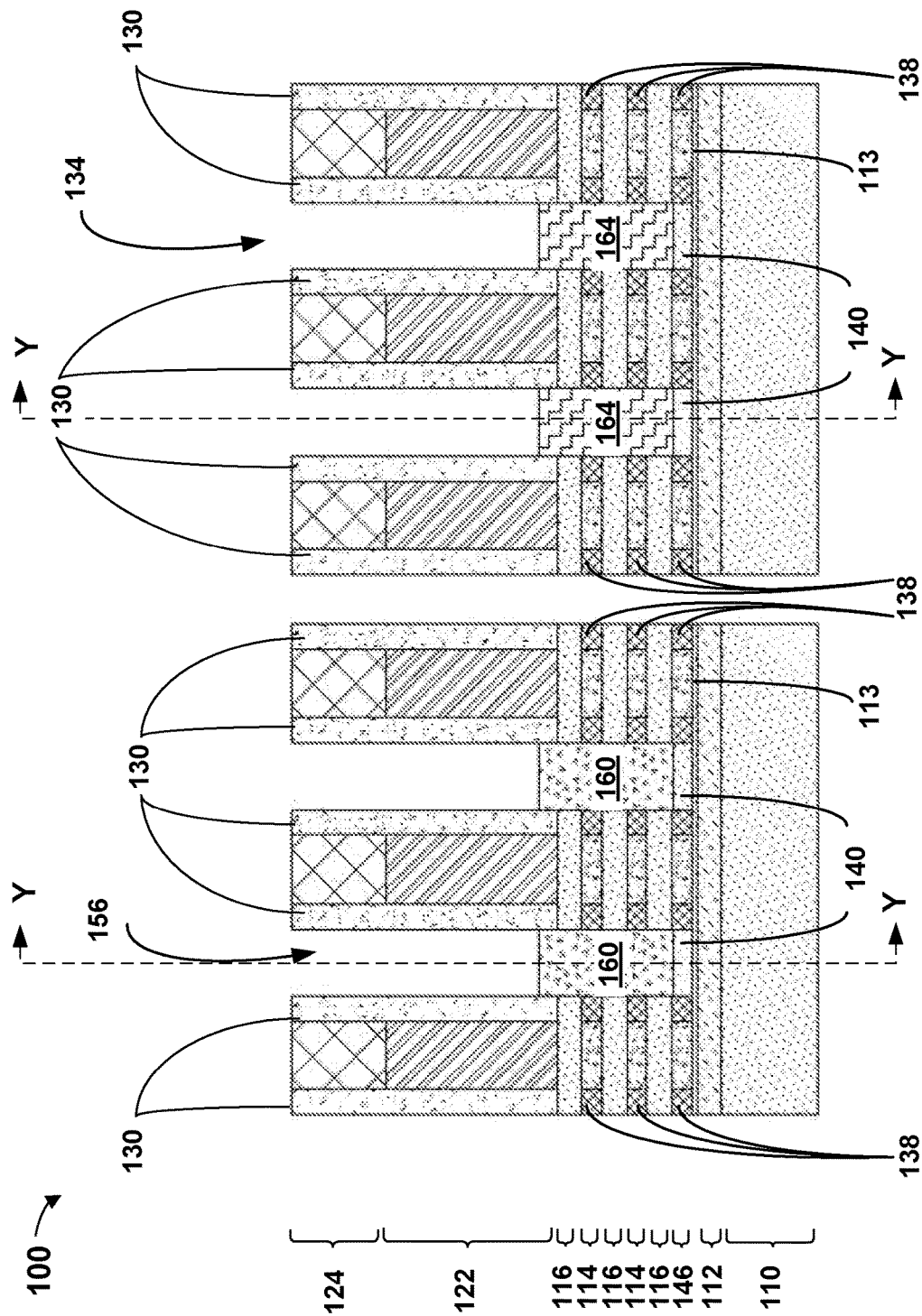


Figure 42
Section X2-X2

Figure 41
Section X1-X1

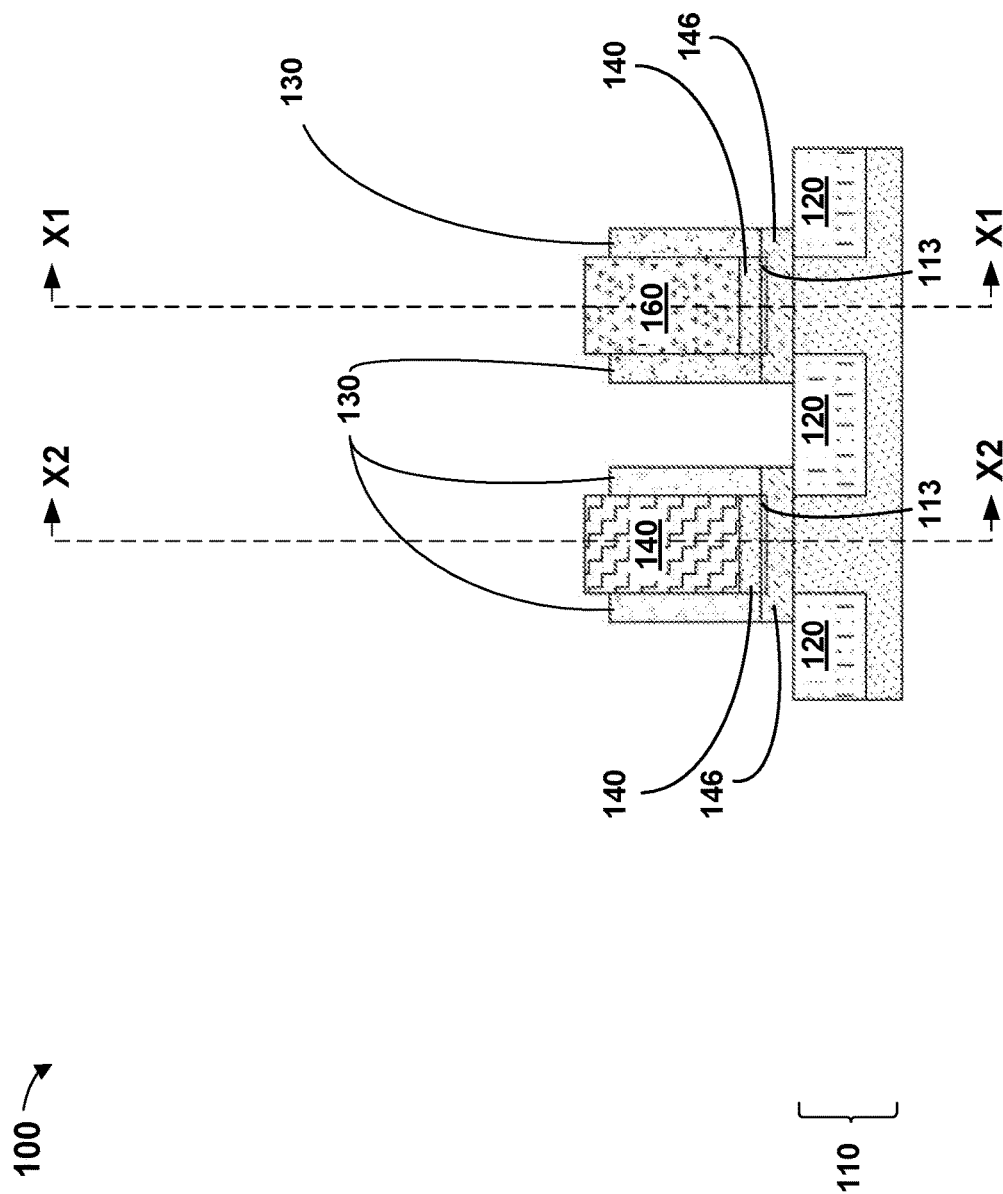


Figure 43
Section Y-Y

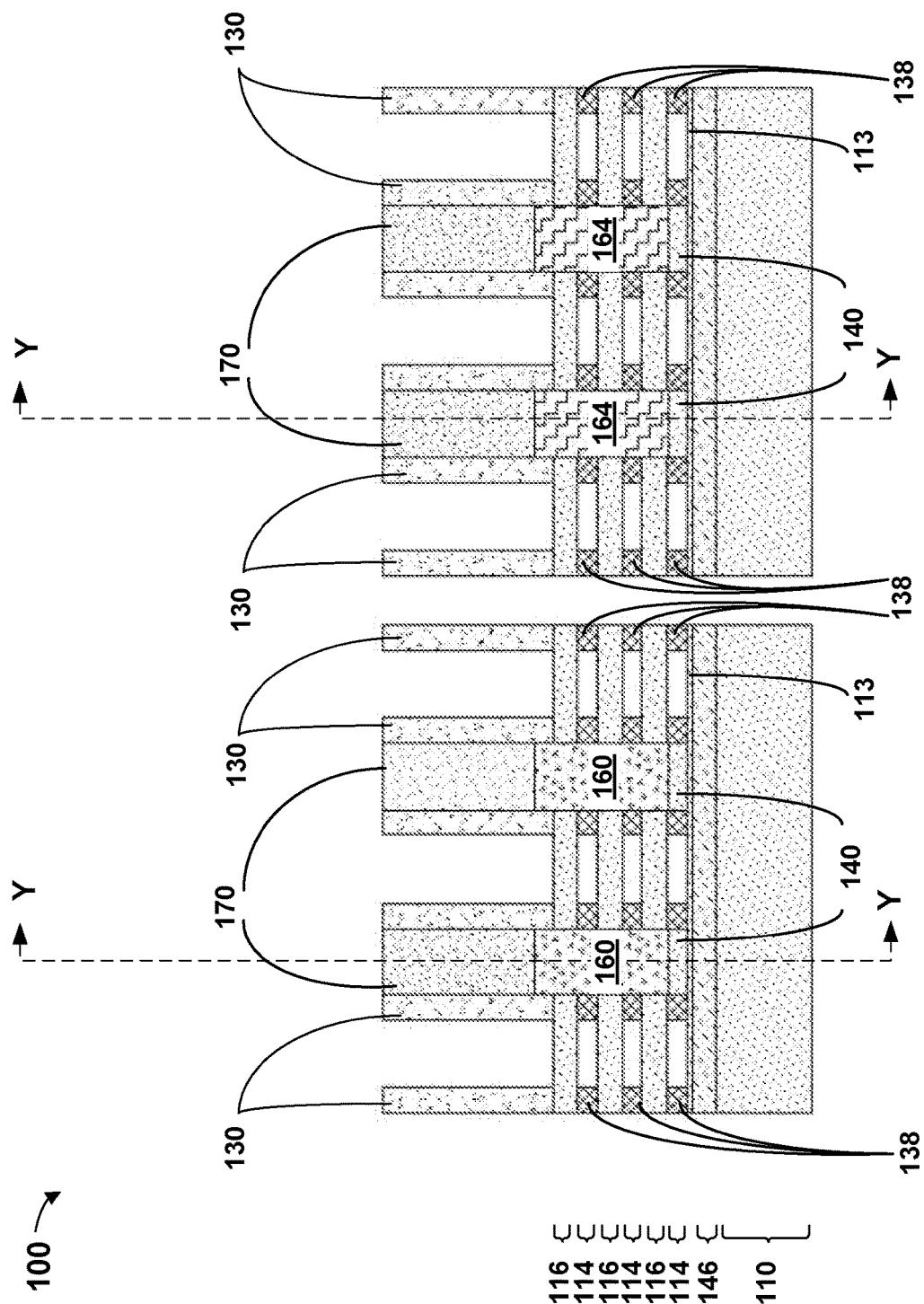


Figure 45
Section X2-X2

Figure 44
Section X1-X1

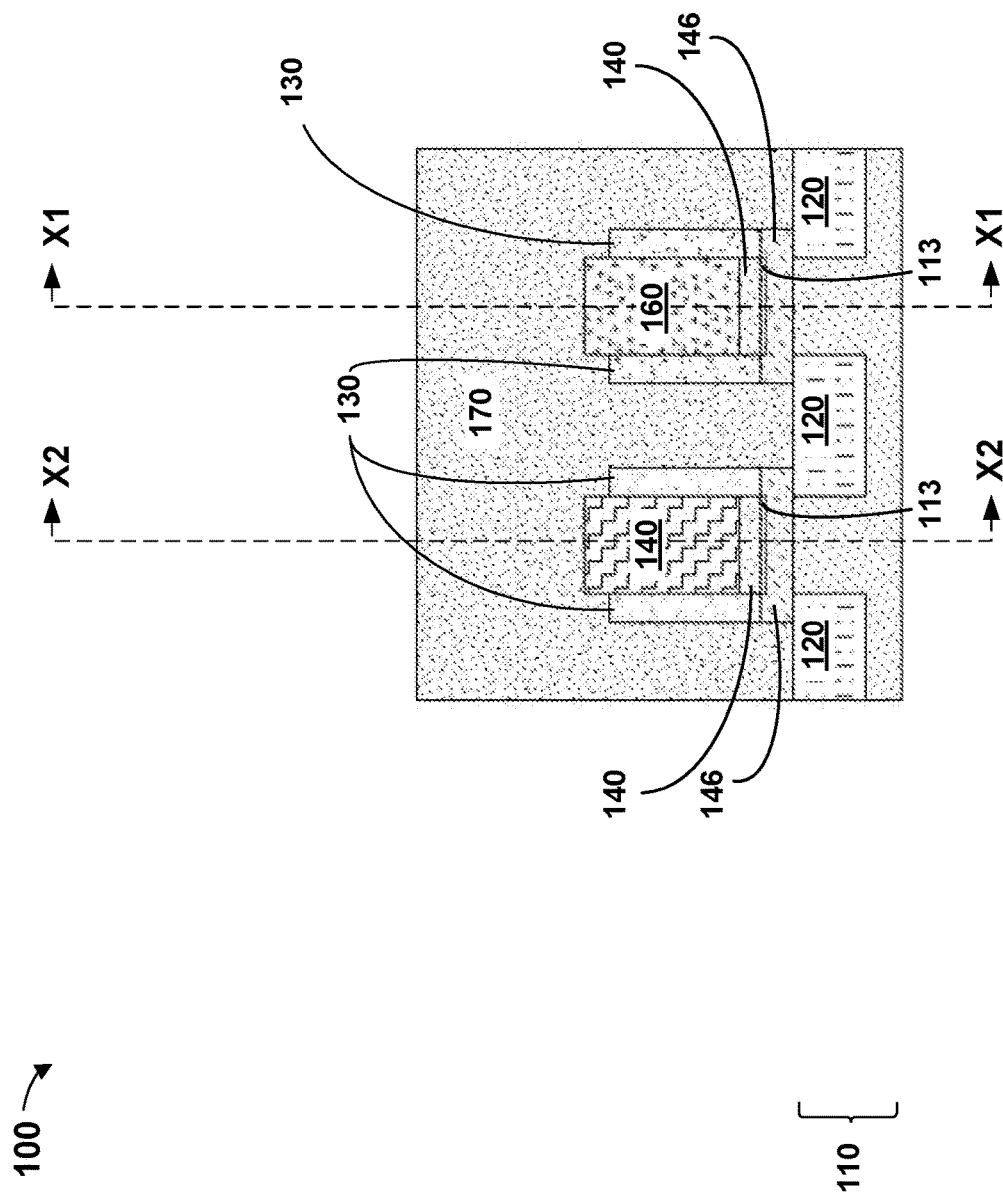


Figure 46
Section Y-Y

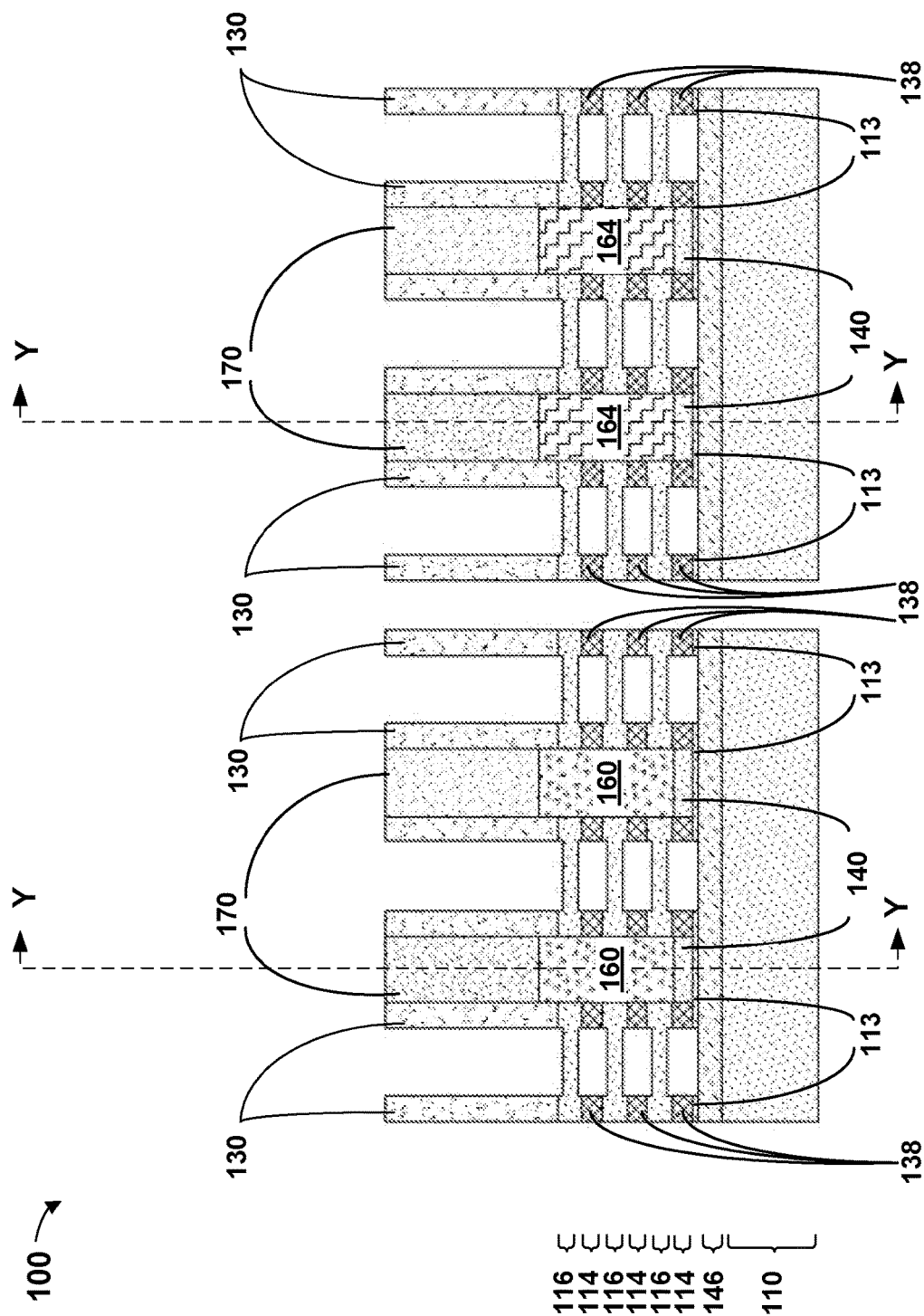


Figure 47

Section X1-X1

Figure 48

Section X2-X2

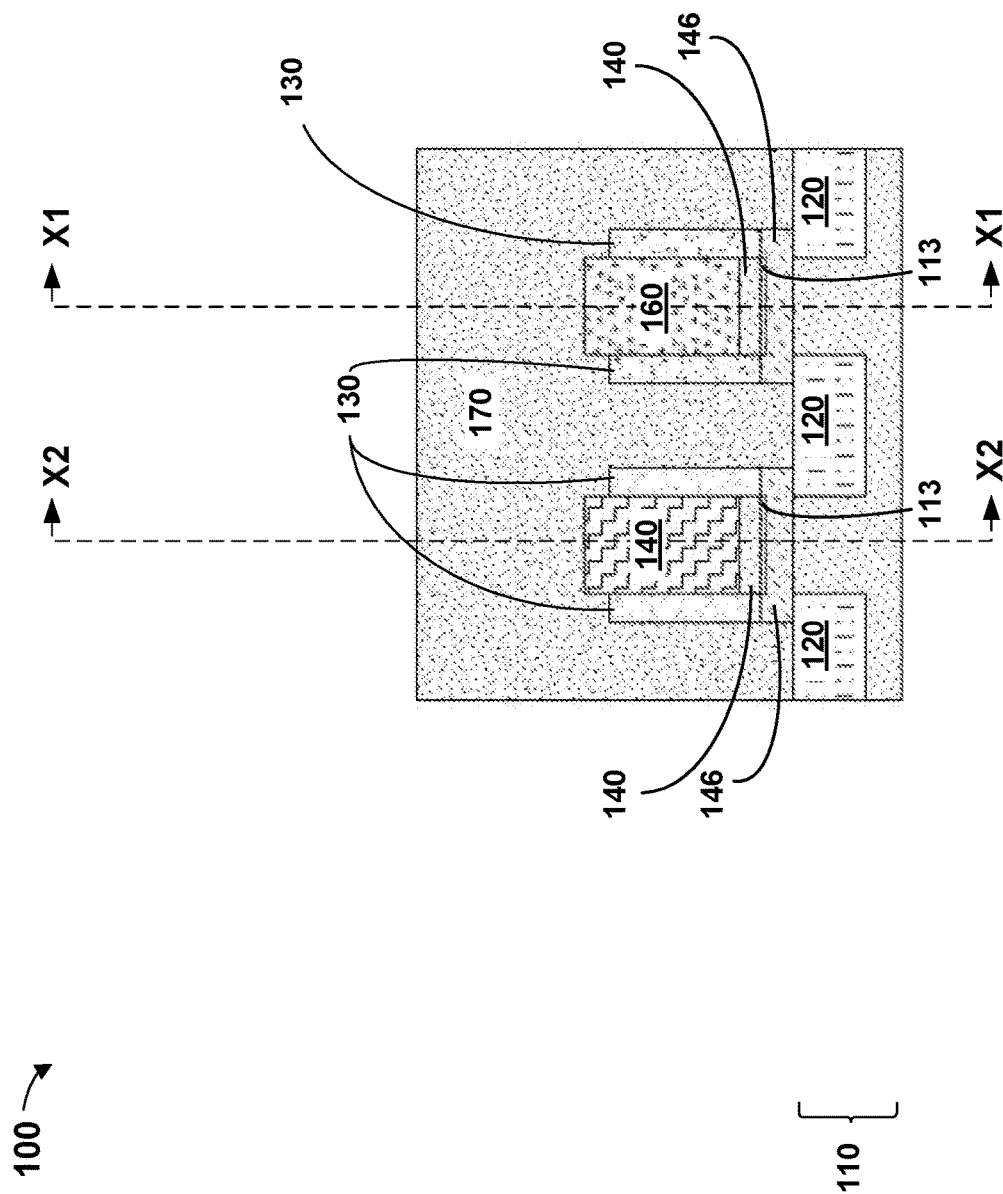


Figure 49
Section Y-Y

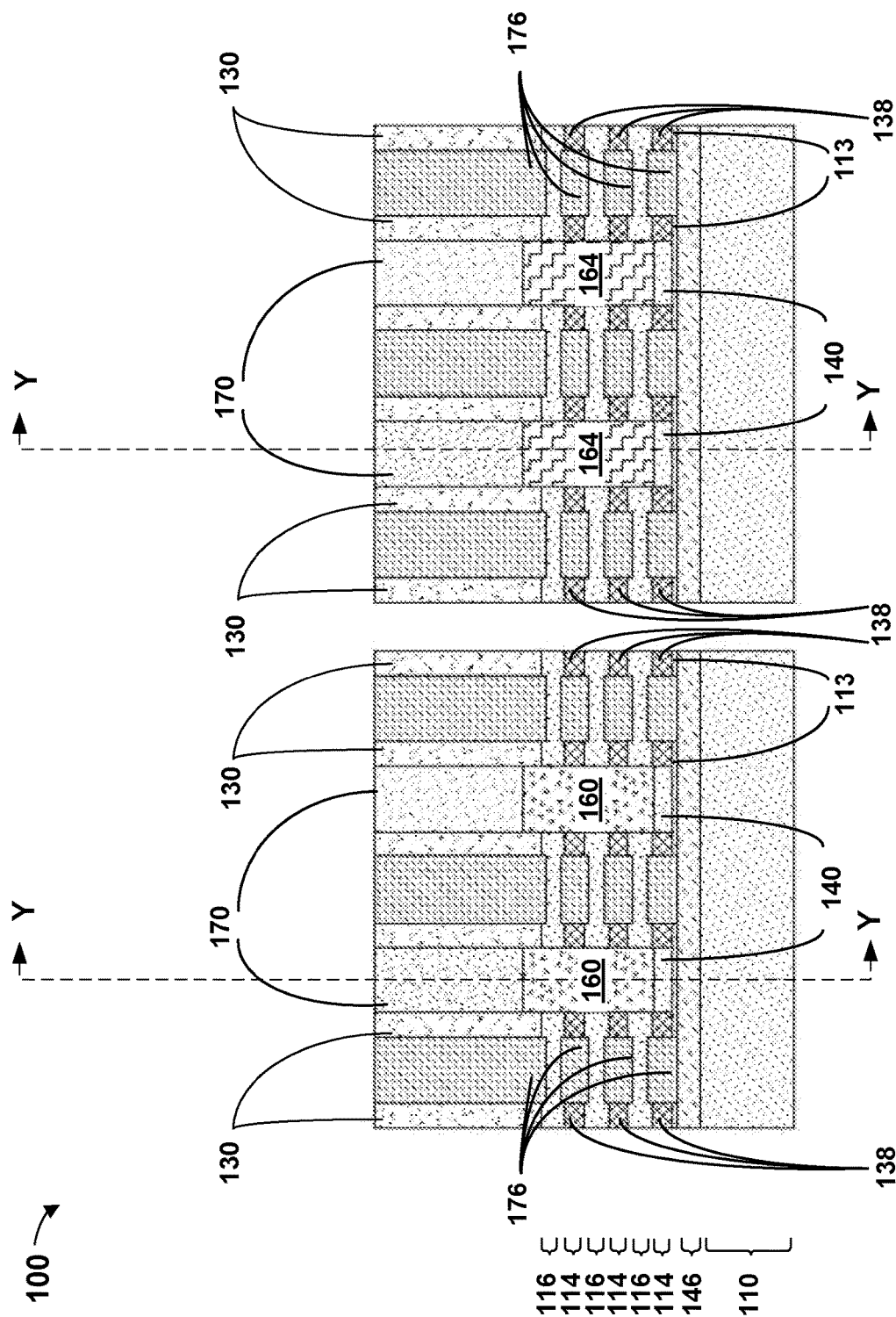


Figure 50

Section X1-X1

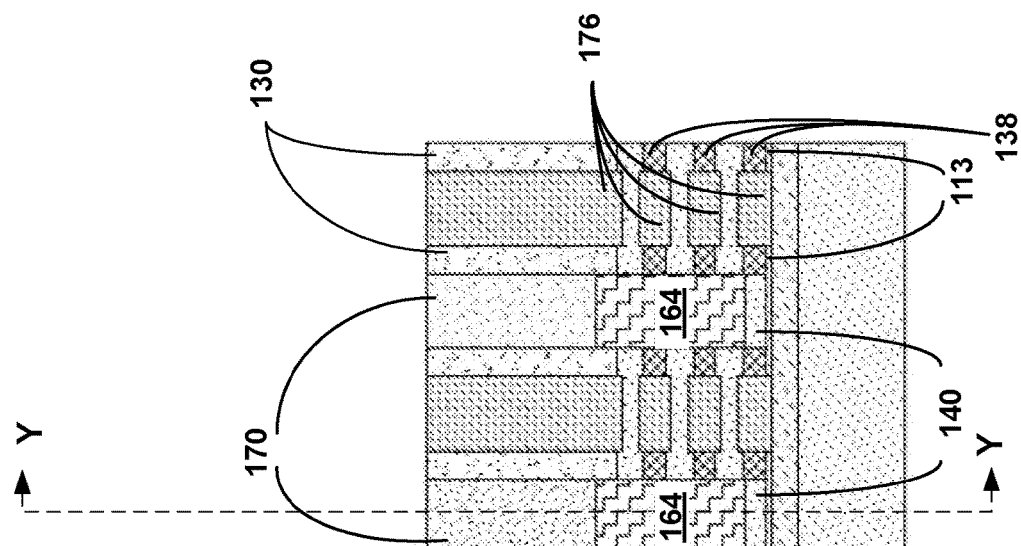


Figure 51

Section X2-X2

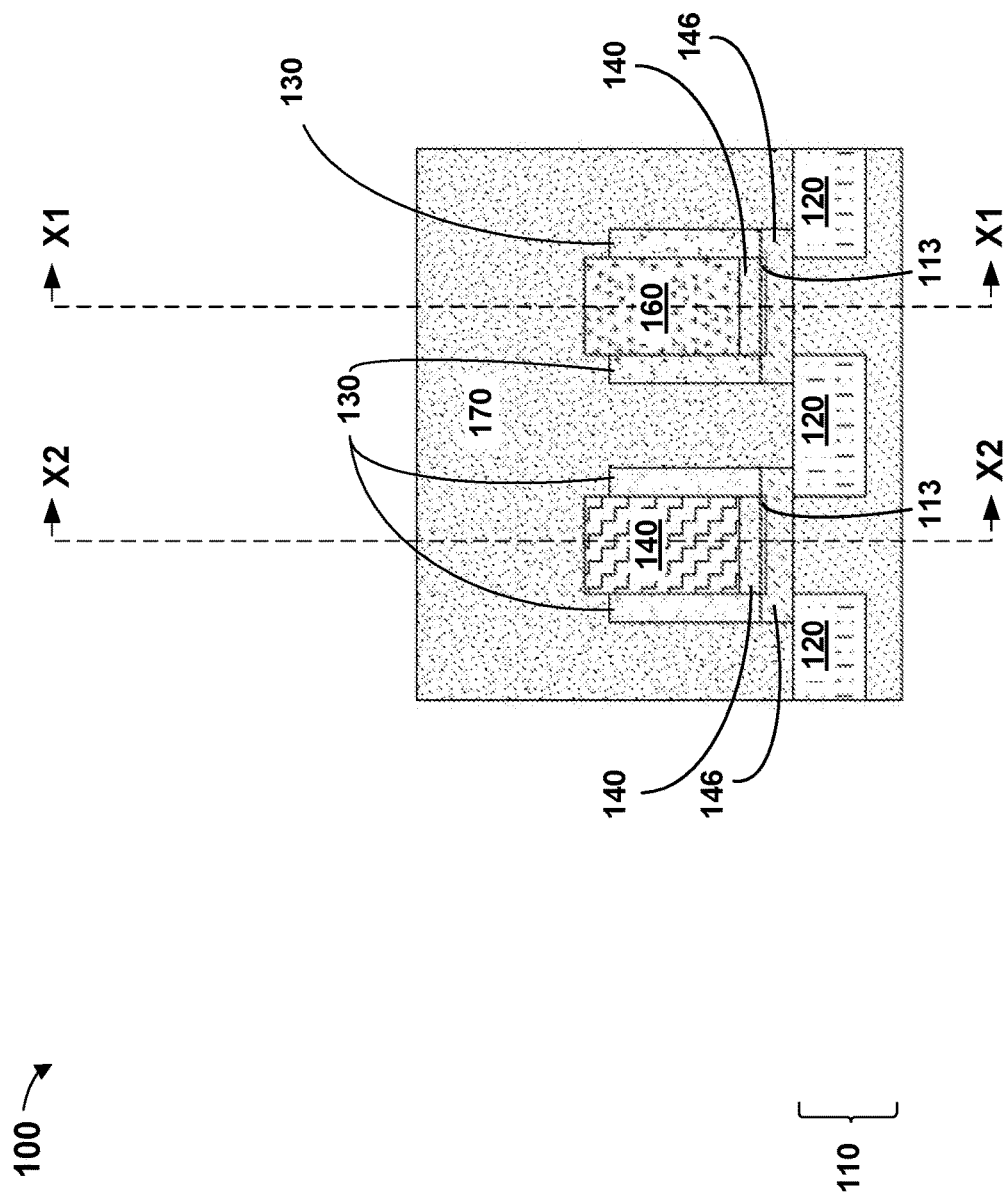


Figure 52
Section Y-Y

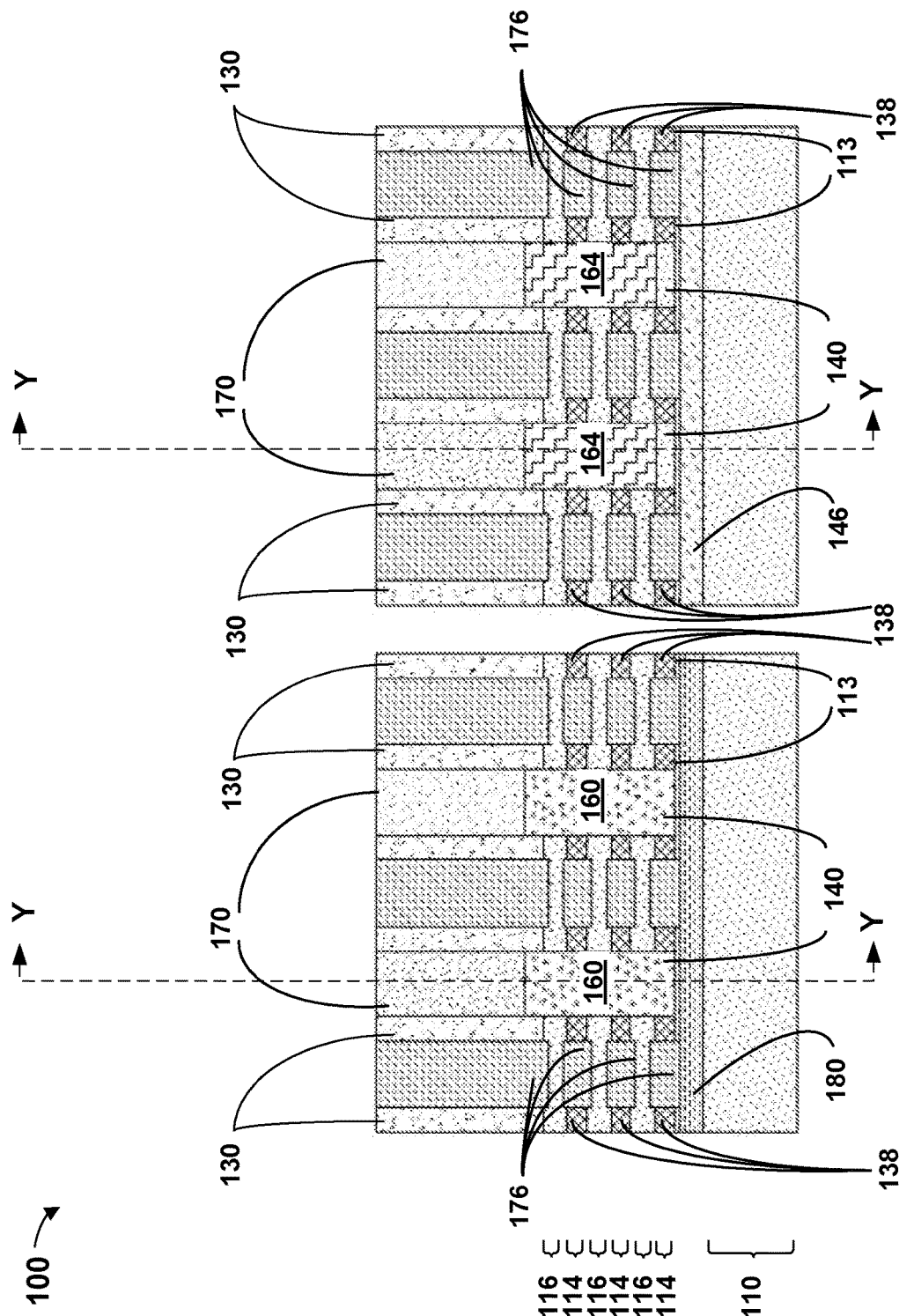


Figure 54
Section X2-X2

Figure 53
Section X1-X1

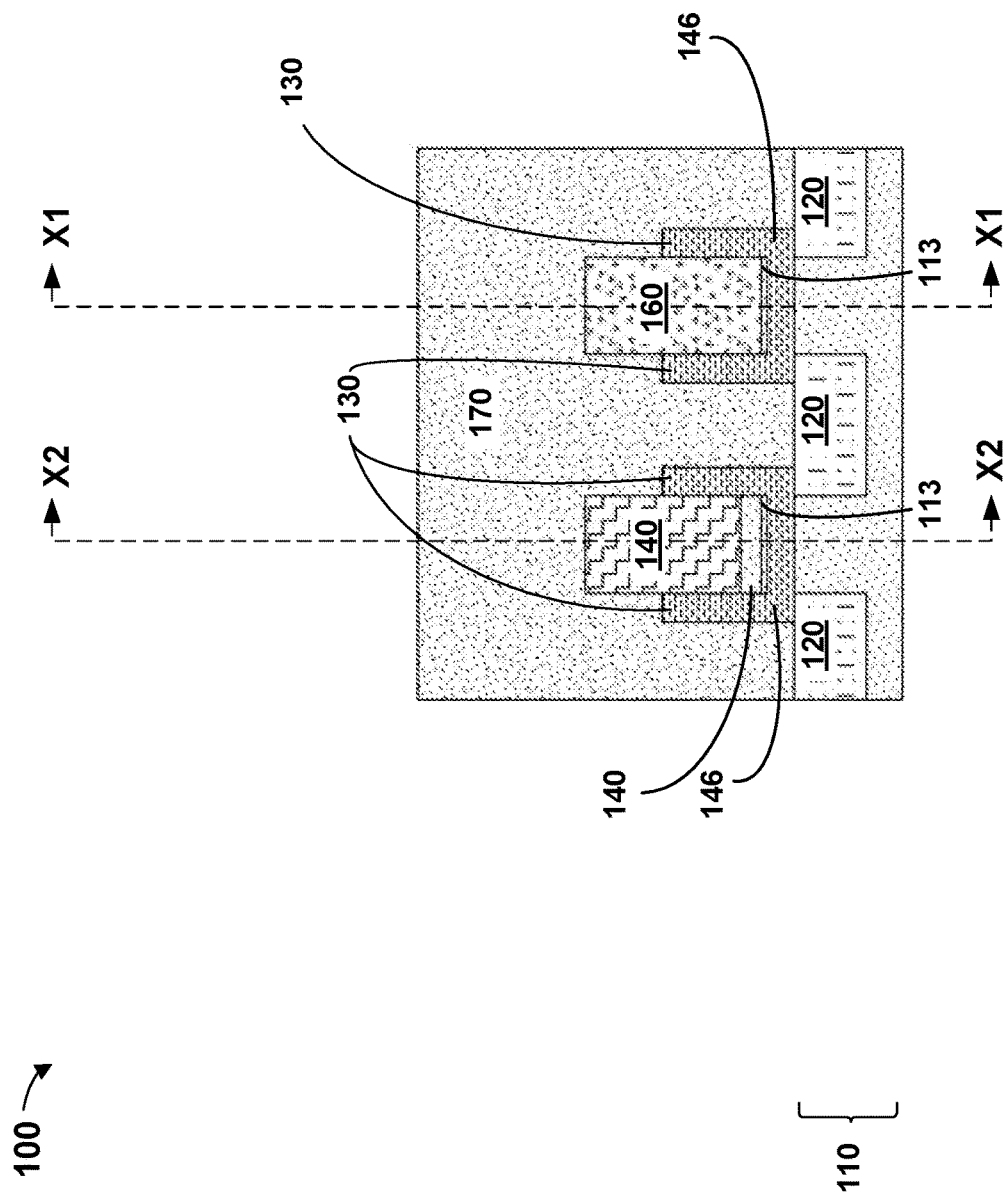


Figure 55
Section Y-Y

NANOSHEET EPITAXY WITH FULL BOTTOM ISOLATION

BACKGROUND

[0001] The present invention relates, generally, to the field of semiconductor manufacturing, and more particularly to fabricating a gate-all-around nanosheet field effect transistor with improved nanosheet epitaxy and full bottom isolation.

[0002] Complementary Metal-oxide-semiconductor (CMOS) technology is commonly used for field effect transistors (hereinafter “FET”) as part of advanced integrated circuits (hereinafter “IC”), such as central processing units (hereinafter “CPUs”), memory, storage devices, and the like. As demands to reduce the dimensions of transistor devices continue, nanosheet FETs help achieve a reduced FET device footprint while maintaining FET device performance. A nanosheet FET includes a plurality of nanosheets extending between a pair of source drain epitaxial regions. The device may be a gate all around transistor in which a gate surrounds at least a portion of the nanosheet channel.

[0003] Bottom dielectric isolation of source and drain epitaxy from the substrate is required for optimal electrical performance as gate lengths approach 12 nanometer and below.

SUMMARY

[0004] According to an embodiment, a gate-all-around device is provided. The gate-all-around device including semiconductor channel layers vertically aligned and stacked one on top of another, the semiconductor channel layers separated from each other by a gate stack material wrapping around the semiconductor channel layers, and a heavily doped p-type field effect transistor (p-FET) source drain epitaxy region adjacent to the semiconductor channel layers, where a horizontal lower surface of the p-FET source drain epitaxy region is adjacent to a horizontal upper surface of an undoped silicon epitaxy.

[0005] According to an embodiment, a semiconductor device is provided. The semiconductor device including a first set of semiconductor channel layers vertically aligned and stacked one on top of another, a heavily doped p-type field effect transistor (p-FET) source drain epitaxy region adjacent to the first set of semiconductor channel layers, where a horizontal lower surface of the p-FET source drain epitaxy region is adjacent to a horizontal upper surface of an undoped silicon epitaxy, and a heavily doped n-type field effect transistor (n-FET) source drain epitaxy doped region adjacent to a second set of semiconductor channel layers, where a horizontal lower surface of the n-FET source-drain epitaxy region is adjacent to the horizontal upper surface of the undoped silicon epitaxy.

[0006] According to an embodiment, a method is provided. The method including forming a first stack of nanosheet layers on a substrate, a second stack of nanosheet layers on the substrate, and a third stack of nanosheet layers on the substrate, the first stack, the second stack and the third stack of nanosheet layers each including alternating layers of a sacrificial and a semiconductor channel vertically aligned and stacked one on top of another, forming a first sacrificial gate across the first stack of nanosheet layers, a second sacrificial gate across the second stack of nanosheet layers and a third sacrificial gate across the third stack of nanosheet layers, and forming an undoped silicon epitaxy between the

first stack and the second stack of nanosheet layers and between the second stack and the third stack of nanosheet layers.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] These and other objects, features and advantages of the present invention will become apparent from the following detailed description of illustrative embodiments thereof, which is to be read in connection with the accompanying drawings. The various features of the drawings are not to scale as the illustrations are for clarity in facilitating one skilled in the art in understanding the invention in conjunction with the detailed description. In the drawings:

[0008] FIG. 1 illustrates a top view of a semiconductor structure at an intermediate stage of fabrication, according to an exemplary embodiment;

[0009] FIGS. 2, 3 and 4 each illustrate a cross-sectional view of the semiconductor structure of FIG. 1 along section lines X1-X1, X2-X2 and Y-Y, respectively, FIGS. 2 and 3 are parallel to each other, and FIG. 4 is perpendicular to FIGS. 2 and 3, according to an exemplary embodiment;

[0010] FIGS. 5, 6 and 7 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates formation of nanosheet stacks, shallow trench isolation and sacrificial gates, according to an exemplary embodiment;

[0011] FIGS. 8, 9 and 10 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates formation of a spacer, according to an exemplary embodiment;

[0012] FIGS. 11, 12 and 13 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates formation of a source drain trench in the nanosheet stack;

[0013] FIGS. 14, 15 and 16 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates formation of inner spacers;

[0014] FIGS. 17, 18 and 19 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates formation of an undoped silicon epitaxy;

[0015] FIGS. 20, 21 and 22 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and recessing a shallow trench isolation region;

[0016] FIGS. 23, 24 and 25 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates removal of a sacrificial epitaxial layer, according to an exemplary embodiment;

[0017] FIGS. 26, 27 and 28 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates formation of a bottom dielectric isolation, according to an exemplary embodiment;

[0018] FIGS. 29, 30 and 31 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and formation of an etch stop liner, according to an exemplary embodiment;

[0019] FIGS. 32, 33 and 34 each illustrate a cross-sectional view of the of the semiconductor FIGS. 32, 33 and 34 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates formation of a first organic planarization layer, according to an exemplary embodiment;

[0020] FIGS. 35, 36 and 37 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates formation of a p-FET source drain trench, according to an exemplary embodiment;

[0021] FIGS. 38, 39 and 40 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates formation of a p-FET source drain epitaxy, according to an exemplary embodiment;

[0022] FIGS. 41, 42 and 43 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates formation of an n-FET source drain epitaxy, according to an exemplary embodiment;

[0023] FIGS. 44, 45 and 46 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates formation of an interlayer dielectric, and removal of gate hard mask, sacrificial gate and sacrificial suspension layers, according to an exemplary embodiment;

[0024] FIGS. 47, 48 and 49 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates selective etching, according to an exemplary embodiment;

[0025] FIGS. 50, 51 and 52 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates formation of a gate stack, according to an exemplary embodiment; and

[0026] FIGS. 53, 54 and 55 each illustrate a cross-sectional view of the of the semiconductor structure along section lines X1-X1, X2-X2 and Y-Y, respectively, and illustrates an alternate embodiment.

[0027] It will be appreciated that for simplicity and clarity of illustration, elements shown in the figures have not necessarily been drawn to scale. For example, the dimensions of some of the elements may be exaggerated relative to other elements for clarity. Further, where considered appropriate, reference numbers may be repeated among the figures to indicate corresponding or analogous features.

DETAILED DESCRIPTION

[0028] Detailed embodiments of the claimed structures and methods are disclosed herein; however, it can be understood that the disclosed embodiments are merely illustrative of the claimed structures and methods that may be embodied in various forms. This invention may, however, be embodied in many different forms and should not be construed as limited to the exemplary embodiments set forth herein. In the description, details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the presented embodiments.

[0029] References in the specification to “one embodiment”, “an embodiment”, “an example embodiment”, etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every

embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

[0030] It will be understood that when an element as a layer, region or substrate is referred to as being “on” or “over” another element, it can be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or “directly over” another element, there are no intervening elements present. It will also be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

[0031] In the interest of not obscuring the presentation of embodiments of the present invention, in the following detailed description, some processing steps or operations that are known in the art may have been combined together for presentation and for illustration purposes and in some instances may have not been described in detail. In other instances, some processing steps or operations that are known in the art may not be described at all. It should be understood that the following description is rather focused on the distinctive features or elements of various embodiments of the present invention.

[0032] A nanosheet field effect transistor (hereinafter “FET”) may be formed from alternating layers of silicon, referred to as channel layers, and silicon germanium, referred to as sacrificial suspension layers, which are then formed into nanosheets. A gate all around structure may be formed on all vertical sides and on a horizontal top surface of a section of the nanosheets. Source-drain epitaxy structures may be formed at the opposite ends of the nanosheet structures.

[0033] The present invention relates, generally, to the field of semiconductor manufacturing, and more particularly to fabricating a nanosheet field effect transistor with full bottom isolation.

[0034] For nanosheet FET devices on a bulk substrate, source drain epitaxy grows from sides of the silicon channel layers of the nanosheet devices and also from the substrate. This may lead to parasitic source-to-drain leakage through the sub-sheet region under the gate region of the nanosheet FET device, which has a negative effect on nanosheet FET performance such as increased parasitic leakage.

[0035] To reduce parasitic source-to-drain leakage between the source and drain epitaxy contact through the substrate, a bottom dielectric isolation (BDI) which isolates the source-drain epitaxy from the substrate is required for optimal electrical performances, especially at effective gate stack lengths of 12 nm and below. The effective gate stack length is defined as the physical distance of the gate stack in the direction of current transport (not accounting for the thickness of the gate dielectric). For example, a gate length of 15 nm may have an effective gate length of 12 nm.

[0036] This results in the source drain epitaxy growing only from the channel layers without a benefit of also

growing from the substrate and makes it difficult to generate strain on the channel regions, or channel layers of the nanosheet.

[0037] Typically, stress can be generated in the source drain epitaxy by having a bottom-up nucleation and growth component from the substrate. However, the addition of a BDI layer may remove this bottom-up nucleation and growth component leading to a fully relaxed source-drain epitaxy, thereby reducing device performance. Here, the inventors propose a new method of introducing stress to the source drain epitaxy while using a BDI layer.

[0038] Stress on the semiconductor channel layer improves performance of the nanosheet FET by generating strain in the channel, improving carrier mobility in the channel and thereby increasing device performance. Strain engineering allows local modification of the band structure in the channel region, affecting the effective mass of the carriers and hence the carrier mobility during transport in the channel region.

[0039] In an embodiment of this invention an undoped silicon epitaxy is formed between stacked nanosheets, a bottom dielectric isolation is formed between the stacked nanosheets, the undoped silicon epitaxy and the substrate. A portion of the undoped silicon epitaxy is removed, and a remaining portions of the undoped silicon epitaxy along with channel regions of the stacked nanosheets are used to grown the source drain epitaxy.

[0040] In an embodiment of this invention, a stacked nanosheet of alternating layers of sacrificial semiconductor material and semiconductor channel material is formed on a substrate over a nanosheet sacrificial layer. The stacked nanosheet is formed into nanosheet stacks and a shallow trench isolation formed between each nanosheet stack. A sacrificial gate and gate hard mask is formed, in a direction perpendicular to the nanosheet stacks. A spacer is formed on opposite sides of the sacrificial gate and gate hard mask, and on opposite sides of the nanosheet stack between sacrificial gates. A source drain trench is formed between adjacent sacrificial gate, gate hard mask and spacers. The source drain trench is formed by removal of vertically aligned portions of the nanosheet stack. In a process of record, the source drain trench may expose the nanosheet sacrificial layer. In this process, as a new feature, portions of the nanosheet stack remain below the source drain trench, where the nanosheet sacrificial layer, a lowermost channel layer and at least a portion of a lowermost sacrificial semiconductor material layer are not removed, and remain between sacrificial gates. A portion of each of the sacrificial suspension layers may be indented, and an inner spacer may be formed on either side of the sacrificial suspension layers. An undoped silicon epitaxy may be formed between nanosheet stacks, and between the spacers. The undoped silicon epitaxy is a new feature. A portion of the shallow trench isolation may be removed, exposing the sacrificial epitaxial layer. The sacrificial epitaxial layer may be removed. A bottom dielectric isolation may be formed where the sacrificial epitaxial layer was removed. This is a unique feature, where the bottom dielectric isolation is under the undoped silicon epitaxy. An etch stop liner may be formed. A first organic planarization layer may be formed and patterned on a portion of the structure. A portion of the etch stop liner and a portion of the undoped epitaxy may be removed where not covered by the first organic planarization layer. A p-FET source drain epitaxy may be formed between sacrificial gates which are

not protected by the first organic planarization layer. The first organic planarization layer and remaining etch stop liner may be removed and the process steps repeated to form an n-FET source drain epitaxy in between sacrificial gates which were originally protected by the first organic planarization layer. Both the p-FET source drain epitaxy and the n-FET source drain epitaxy may be grown from both the undoped silicon epitaxy and from the channel layers. An interlayer dielectric may be formed between adjacent sacrificial gates, gate hard mask and spacers. The gate hard mask, the sacrificial gate, and the sacrificial suspension layers may be removed. Selective etching may be performed. A gate stack may be formed where the gate hard mask, the sacrificial gate, and the sacrificial suspension layers were removed. An alternate embodiment results in the p-FET source drain epitaxy directly contacting a second bottom dielectric isolation.

[0041] Embodiments of the present invention disclose a structure and a method of forming a gate-all-around nanosheet with source drain epitaxy formed from undoped silicon epitaxy and from channel regions of a stacked nanosheet, and are described in detail below by referring to the accompanying drawings, in accordance with an illustrative embodiment.

[0042] Referring now to FIG. 1, a semiconductor structure **100** (hereinafter “structure”) at an intermediate stage of fabrication is shown according to an exemplary embodiment. FIG. 1 is a top view of the structure **100** at an intermediate stage of fabrication. The structure **100** of FIG. 1 may be formed or provided.

[0043] Section line Y-Y of FIG. 1 is parallel with subsequently formed gate lines and is centered between adjacent gates in a source drain epitaxy region. Section line X1-X1 and section line X2-X2 of FIG. 1 are parallel to each other, are an intersecting line along a length of fins or nanosheet stacks of the structure **100** and are each perpendicular to section line Y-Y.

[0044] Referring now to FIGS. 2, 3 and 4, the structure **100** is shown according to an exemplary embodiment. FIGS. 2 and 3 are each a cross-sectional view of the structure **100** along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. 4 is a cross-sectional view of the structure **100** along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2.

[0045] The structure **100** may include a substrate **110**. The substrate **110** may be, for example, a bulk substrate, which may be made from any of several known semiconductor materials such as, for example, silicon, germanium, silicon-germanium alloy, and compound (e.g. III-V and II-VI) semiconductor materials. Non-limiting examples of compound semiconductor materials include gallium arsenide, indium arsenide, and indium phosphide, or indium gallium arsenide. Typically, the substrate **110** may be approximately, but is not limited to, several hundred microns thick. In other embodiments, the substrate **110** may be a layered semiconductor such as a silicon-on-insulator or SiGe-on-insulator, where a buried insulator layer, separates a base substrate from a top semiconductor layer.

[0046] A stacked nanosheet of alternating layers of sacrificial semiconductor material and semiconductor channel material stacked one on top of another, may be formed on the substrate **110**. It should be noted that, while a limited number of alternating layers are depicted, any number of alternating layers may be formed.

[0047] The alternating layers of sacrificial semiconductor material and semiconductor channel material may include a nanosheet sacrificial epitaxial layer 112 (hereinafter “sacrificial epitaxial layer”) on the substrate 110, covered by a semiconductor seed material layer 113 (hereinafter “seed layer”), covered by a sacrificial suspension semiconductor material layer 114 (hereinafter “sacrificial suspension layer”), covered by a semiconductor channel material layer 116 (hereinafter “channel layer”), covered by a sacrificial suspension layer 114, covered by a channel layer 116, covered by a sacrificial suspension layer 114, covered by a channel layer 116.

[0048] The sacrificial epitaxial layer 112, may, for example, be silicon germanium with a germanium concentration between 45 and 70 atomic percent, although percentages greater than 70 percent and less than 45 percent may be used. The sacrificial epitaxial layer 112 can be formed using known deposition techniques or an epitaxial growth technique as described below. The sacrificial epitaxial layer 112 will subsequently be removed selective to the remaining alternating layers, as described below.

[0049] Each sacrificial suspension layer 114 is composed of a first semiconductor material which differs in composition from at least an upper portion of the substrate 110, the channel layers 116 and the sacrificial epitaxial layer 112. In an embodiment, each sacrificial suspension layer 114 may be a silicon-germanium semiconductor alloy and have a germanium concentration less than 50 atomic percent. In another example, each sacrificial suspension layer 114 may have a germanium concentration ranging from about 15 atomic percent to about 35 atomic percent, although percentages greater than 35 percent and less than 15 percent may be used. Each sacrificial suspension layer 114 can be formed using known deposition techniques or an epitaxial growth technique as described below.

[0050] Each channel layer 116 is composed of a second semiconductor material which differs in composition from at least the upper portion of the substrate 110, the sacrificial suspension layers 114 and the sacrificial epitaxial layer 112. Each channel layer 116 has a different etch rate than the first semiconductor material of sacrificial suspension layer 114 and has a different etch rate than the sacrificial epitaxial layer 112. The sacrificial suspension layer 114 has a different etch rate than the sacrificial epitaxial layer 112. The second semiconductor material can be, for example, silicon. The second semiconductor material, for each channel layer 116, can be formed using known deposition techniques or an epitaxial growth technique as described above.

[0051] In an embodiment, the seed layer 113 may be composed of the second semiconductor material.

[0052] In an embodiment, the sacrificial epitaxial layer 112 may be the same material as the sacrificial suspension layer 114, and would have the same etch rate as each other.

[0053] The sacrificial epitaxial layer 112, the seed layer and the alternating layers of sacrificial suspension layers 114 and the channel layers 116 can be formed by sequential epitaxial growth of the sacrificial epitaxial layer material and alternating layers of the first semiconductor material and the second semiconductor material.

[0054] The terms “epitaxially growing and/or depositing” and “epitaxially grown and/or deposited” mean the growth of a semiconductor material on a deposition surface of a semiconductor material, in which the semiconductor material being grown has the same crystalline characteristics as

the semiconductor material of the deposition surface. In an epitaxial deposition technique, the chemical reactants provided by the source gases are controlled and the system parameters are set so that the depositing atoms arrive at the deposition surface of the semiconductor substrate with sufficient energy to move around on the surface and orient themselves to the crystal arrangement of the atoms of the deposition surface. Therefore, an epitaxial semiconductor material has the same crystalline characteristics as the deposition surface on which it is formed.

[0055] Examples of various epitaxial growth techniques include, for example, rapid thermal chemical vapor deposition (RTCVD), low-energy plasma deposition (LEPD), ultra-high vacuum chemical vapor deposition (UHV-CVD), low pressure chemical vapor deposition (LPCVD), atmospheric pressure chemical vapor deposition (APCVD) and molecular beam epitaxy (MBE). The temperature for epitaxial deposition typically ranges from approximately 550° C. to approximately 900° C. Although higher temperature typically results in faster deposition, the faster deposition may result in crystal defects and film cracking. The epitaxial growth of first and second semiconductor materials that provide the sacrificial epitaxial layer 112, the sacrificial suspension layers 114 and the channel layers 116, respectively, can be performed utilizing any well-known precursor gas or gas mixture. Carrier gases like hydrogen, nitrogen, helium and argon can be used.

[0056] The sacrificial epitaxial layer 112 may have a thickness ranging from about 5 nm to about 15 nm. The seed layer 113 may have a thickness ranging from about 2 nm to about 5 nm. The sacrificial suspension layers 114 may each have a thickness ranging from about 5 nm to about 12 nm, while the channel layers 116 may each have a thickness ranging from about 5 nm to about 12 nm. Each sacrificial suspension layer 114 may have a thickness that is the same as, or different from, a thickness of each channel layer 116. In an embodiment, each sacrificial suspension layer 114 has an identical thickness. In an embodiment, each channel layer 116 has an identical thickness.

[0057] Referring now to FIGS. 5, 6 and 7, the structure 100 is shown according to an exemplary embodiment. FIGS. 5 and 6 are each a cross-sectional view of the structure 100 along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. 7 is a cross-sectional view of the structure 100 along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. The stacked nanosheet may be formed into nanosheet stacks. A shallow trench isolation region (hereinafter “STI”) 120 may be formed between each nanosheet stack. A sacrificial gate 122 and a gate hard mask 124 may be formed.

[0058] As shown in FIG. 7, the stacked nanosheet may be formed into nanosheet stacks. The nanosheet stacks may have a length perpendicular to section line Y-Y and parallel to section lines X1-X1 and X2-X2. The nanosheet stacks may be formed by methods known in the arts, and include steps such as forming a nanosheet hard mask (not shown) on the stacked nanosheet, patterning the nanosheet hard mask (not shown) and subsequent formation of one or more trenches 118, by removal of portions of each layer of the stacked nanosheet. The trench 118 may be formed between each nanosheet stack by an anisotropic etching technique, such as, for example, reactive ion etching (RIE), and stopping on etching a portion of the substrate 110 for subsequent

formation of the STI 120 between each nanosheet stack. The nanosheet hard mask (not shown) may be removed.

[0059] Each nanosheet stack may include the sacrificial epitaxial layer 112 covered by the seed layer 113, covered by the alternating layers of channel layers 116 and sacrificial suspension layers 114. In this embodiment, and only by way of an example, the nanosheet stack includes three layers of channel layers 116 alternating with three layers of the sacrificial suspension layers 114. The material stacks that can be employed in embodiments of the present invention are not limited to the specific embodiment illustrated in the Figures. There may be any number of nanosheet stacks on the structure 100.

[0060] The nanosheet stack is used to produce a gate all around device that includes vertically stacked semiconductor channel material nanosheets for a p-FET or an n-FET device.

[0061] The STI 120 is formed in a portion of the trench 118. The STI 120 may be formed between adjacent nanosheet stacks in the portion of the trench 118, between adjacent nanosheet stacks. The STI 120 may be a dielectric material and may be between adjacent nanosheet stacks and formed using known deposition, planarization and etching techniques. Adjacent nanosheet stacks are isolated from one another in the substrate 110 by the STI 120. A lower horizontal surface and a portion of a vertical side surface of the STI 120 may be adjacent to a lower horizontal surface and a vertical side surface of the substrate 110. The lower horizontal surface of the STI 120 may be below a lower horizontal surface of the stack sacrificial layer 112. An upper horizontal surface of the STI 120 may be at a same level or higher than an upper horizontal surface of the seed layer 113.

[0062] The sacrificial gate 122 may include a single sacrificial material or a stack of one or more sacrificial materials. The at least one sacrificial material can be formed by forming a blanket layer (or layers) of a material (or various materials) and then patterning the material (or various materials) by lithography and an etch. The sacrificial gate 122 can include any material including, for example, polysilicon, amorphous silicon, or multilayered combinations thereof. The sacrificial gate 122 can be formed using any deposition technique including, for example, chemical vapor deposition (CVD), physical vapor deposition (PVD), high density plasma (HDP) deposition, and spin on techniques. Optionally, the gate hard mask 124 may be formed as part of the sacrificial gate 122 in accordance with known techniques. As shown in FIGS. 5 and 6, there are two sacrificial gates 122. In an embodiment, there may be any number of sacrificial gates 122 formed.

[0063] In an embodiment, the sacrificial gate 122 is deposited with a thickness sufficient to fill, or substantially fill, the spaces between adjacent nanosheet stacks (not shown) and cover an upper surface and exposed side surfaces of an upper surface of the uppermost sacrificial channel layer 116 of the nanosheet stack. The sacrificial gate 122 may be adjacent to vertical side surfaces of the nanosheet stack, including vertical side surfaces of the sacrificial epitaxial layer 112, the channel layers 116 and the sacrificial suspension layers 114. A height of the sacrificial gate 122 may be much thicker than the underlying structure and may have a height between 50 nm and 100 nm above the nanosheet stack.

[0064] The gate hard mask 124 may be formed over a horizontal upper surface of the sacrificial gate 122, by

methods known in the art. The gate hard mask 124 may have vertical side surfaces which align with vertical side surfaces of the sacrificial gate 122.

[0065] Referring now to FIGS. 8, 9 and 10, the structure 100 is shown according to an exemplary embodiment. FIGS. 8 and 9 are each a cross-sectional view of the structure 100 along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. 10 is a cross-sectional view of the structure 100 along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. A spacer 130 may be formed.

[0066] The spacer 130 may be formed on either side of the sacrificial gate 122 and the gate hard mask 124 and on either side of each nanosheet stack. The spacer 130 may be formed on opposite sides of the nanosheet stack between sacrificial gates 122, as shown in FIG. 10.

[0067] The spacer 130 may be formed after several processes, including for example, conformally depositing or growing a dielectric and performing an anisotropic etch process. The spacer 130 may include any dielectric material such as a low-k material such as SiON, SiOCN, SiOC and SiBCN, among other materials, and may include a single layer or may include multiple layers of dielectric material.

[0068] Referring now to FIGS. 11, 12 and 13, the structure 100 is shown according to an exemplary embodiment. FIGS. 11 and 12 are each a cross-sectional view of the structure 100 along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. 13 is a cross-sectional view of the structure 100 along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. A source drain trench 134 may be formed by recessing a vertical portion of the nanosheet stack.

[0069] The source drain trench 134 may be formed by an anisotropic etching technique, such as, for example, reactive ion etching (RIE), and etching continues until a portion of the lowermost sacrificial suspension layer 114 of the nanosheet stacks is removed. A lowermost surface of the trench 134 may be below a top surface of the lowermost sacrificial suspension layer 114. The anisotropic etching may remove aligned vertical portions of the stacked nanosheet stack between adjacent sacrificial gates 122, gate hard mask 124 and spacer 130. The sacrificial gate 122, the gate hard mask 124 and the spacers 130 may protect remaining portions of the nanosheet stack. The seed layer 113 and the sacrificial epitaxial layer 112 below the nanosheet stack may remain intact. The vertical portion of the nanosheet stack may be recessed for subsequent formation of a source drain epitaxy in the source drain trench 134. The hard mask (not shown) may be removed.

[0070] A resulting structure has the nanosheet stacks remaining as protected by the combined structure of the sacrificial gate 122, gate hard mask 124 and spacer 130, and between these structures a remaining portion of the lowermost sacrificial suspension layer 114 above the seed layer 113 of the nanosheet stacks, and the sacrificial epitaxial layer 112 below the seed layer 113 of the nanosheet stacks remains.

[0071] An upper horizontal surface of the remaining portion of the lowermost sacrificial suspension layer 114 may form a lower horizontal surface of the source drain trench 134. Vertical side surfaces of two adjacent spacers 130 may form two opposite sidewalls of the source drain trench 134. Vertical side surfaces of the alternating sacrificial suspension

layers 114 and the channel layers 116 of the nanosheet stack may form the remaining two opposite sidewalls of the source drain trench 134.

[0072] Referring now to FIGS. 14, 15 and 16, the structure 100 is shown according to an exemplary embodiment. FIGS. 14 and 15 are each a cross-sectional view of the structure 100 along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. 16 is a cross-sectional view of the structure 100 along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. A portion of each of the sacrificial suspension layers 114 may be indented, and an inner spacer 138 may be formed on either side of the sacrificial suspension layers 114.

[0073] A portion of each of the sacrificial suspension layers 114 on either side of where the vertical portion of the nanosheet stack was recessed, may be indented. The portion of each of the sacrificial suspension layers 114 may be indented selective to the sacrificial gate 122, the gate hard mask 124, the spacer 130, the channel layers 116, the sacrificial epitaxial layer 112 and the STI 120, using methods known in the arts.

[0074] While removing the portion of each of the sacrificial suspension layers 114, a remaining portion of the lowermost sacrificial suspension layers 114 which had remained between adjacent nanosheet stacks may be removed, resulting in an exposed upper surface of the seed layer 113.

[0075] The inner spacer 138 may be formed on either side of the sacrificial suspension layers 114. Outer vertical sides of the inner spacer 138 may vertically align with the channel layers 116 and inner vertical sides of the inner spacer 138 may vertically align with remaining portions of the sacrificial suspension layers 114.

[0076] The inner spacers 138 may be formed after several processes, including for example, conformally depositing or growing a dielectric and performing an isotropic etch process. The inner spacer 138 may include any dielectric material such as silicon nitride and may include a single layer or may include multiple layers of dielectric material.

[0077] Referring now to FIGS. 17, 18 and 19, the structure 100 is shown according to an exemplary embodiment. FIGS. 17 and 18 are each a cross-sectional view of the structure 100 along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. 19 is a cross-sectional view of the structure 100 along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. An undoped silicon epitaxy 140 may be formed.

[0078] The undoped silicon epitaxy 140 may be formed where the vertical portion of each nanosheet stack was removed, in the source drain trench 134, and may be partially recessed using methods known in the art.

[0079] The undoped silicon epitaxy 140 may be epitaxially grown, as described above, in a region between adjacent sacrificial gates 122, the gate hard mask 124 and spacers 130, where the vertical portion of the nanosheet stack was recessed. The undoped silicon epitaxy 140 may be in direct contact with end portions of the channel layers 116 of the nanosheet stack and end portions of the inner spacer 138 surrounding the sacrificial suspension layers 114. The undoped silicon epitaxy 140 may be grown from the seed layer 113 at a lower surface of the source drain trenches 134, as well as grown from the channel layers 116.

[0080] In this invention, the undoped silicon epitaxy 140 may be grown from the seed layer 113 and the channel layers

116 allowing for a lower defectivity, a higher strain and uniform growth of the undoped silicon epitaxy 140, allowing for an improved performance, while also have a sacrificial epitaxial layer 112 protecting the nanosheet FETs from the substrate 110.

[0081] An upper surface of the undoped silicon epitaxy 140 may be at a higher level than the nanosheet stack between adjacent sacrificial gates 122, gate hard mask 124 and spacers 130.

[0082] Referring now to FIGS. 20, 21 and 22, the structure 100 is shown according to an exemplary embodiment. FIGS. 20 and 21 are each a cross-sectional view of the structure 100 along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. 22 is a cross-sectional view of the structure 100 along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. A portion of the spacer 130 may be recessed to expose a vertical side surface of the sacrificial epitaxial layer 112.

[0083] The portion of the STI 140 may be removed selective to the undoped silicon epitaxy 140, the sacrificial gate 122, the gate hard mask 124, the spacer 130, the channel layers 116, the inner spacers 138, the sacrificial suspension layers 114, the seed layer 113, the sacrificial epitaxial layer 112 and the STI 120, using methods known in the arts. A vertical side surface and a lower surface of the seed layer 113 may be exposed when the portion of the STI 140 is removed. A vertical side surface of the sacrificial epitaxial layer 112 is exposed when the portion of the STI 140 is removed.

[0084] Referring now to FIGS. 23, 24 and 25, the structure 100 is shown according to an exemplary embodiment. FIGS. 23 and 24 are each a cross-sectional view of the structure 100 along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. 25 is a cross-sectional view of the structure 100 along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. The sacrificial epitaxial layer 112 may be removed, forming a gap 144.

[0085] The sacrificial epitaxial layer 112 may be removed selective to the seed layer 113, the sacrificial suspension layers 114, the channel layers 116, the substrate 110, the STI 120, the sacrificial gate 122, the gate hard mask 124, the spacer 130 and the undoped silicon epitaxy 140. For example, a dry etching technique can be used to selectively remove the sacrificial epitaxial layer 112, such as, for example, using vapor phase HCl dry etch.

[0086] Referring now to FIGS. 26, 27 and 28, the structure 100 is shown according to an exemplary embodiment. FIGS. 26 and 27 are each a cross-sectional view of the structure 100 along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. 28 is a cross-sectional view of the structure 100 along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. A bottom dielectric isolation 146 may be formed in the gap 144.

[0087] The bottom dielectric isolation 146 may be formed where the sacrificial epitaxial layer 112 has been removed. The bottom dielectric isolation 146 may be formed between the seed layer 113 of the nanosheet stack and the substrate 110. Vertical side surfaces of the bottom dielectric isolation 146 may align with vertical side surfaces of the spacer 130. A lower horizontal surface of the bottom dielectric isolation 146 may cover a portion of the STI 120.

[0088] The bottom dielectric isolation 146 may be formed after several processes, including for example, conformally depositing or growing a dielectric and performing an aniso-

tropic etch process. The bottom dielectric isolation **146** may include any dielectric material such as silicon nitride and may include a single layer or may include multiple layers of dielectric material.

[0089] Referring now to FIGS. **29**, **30** and **31**, the structure **100** is shown according to an exemplary embodiment. FIGS. **29** and **30** are each a cross-sectional view of the structure **100** along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. **31** is a cross-sectional view of the structure **100** along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. An etch stop liner **150** may be formed on the structure **100**.

[0090] The etch stop liner **150** may be conformally formed on the structure **100**. The etch stop liner **150** may cover an upper surface of the undoped silicon epitaxy **140**, an upper surface and outer side surface of the spacer **130**, a vertical side surface of the bottom dielectric isolation **146**, an upper surface of the STI **120**, and an upper surface of the gate hard mask **124**. The etch stop liner **150** may be deposited using typical deposition techniques, for example, atomic layer deposition (ALD), molecular layer deposition (MLD), chemical vapor deposition (CVD), physical vapor deposition (PVD), high density plasma (HDP) deposition, and spin on techniques, followed by an anisotropic vertical etch process such as a reactive ion etch (RIE), or any suitable etch process. In an embodiment, the etch stop liner **150** may include one or more layers. The etch stop liner **150** may include materials such as a nitride, a silicon nitride and an oxide. The etch stop liner **150** may be referred to as a contact etch stop layer (CESL) **150**. The etch stop liner **150** may be 4 nm thick.

[0091] Referring now to FIGS. **32**, **33** and **34**, the structure **100** is shown according to an exemplary embodiment. FIGS. **32** and **33** are each a cross-sectional view of the structure **100** along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. **34** is a cross-sectional view of the structure **100** along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. A first organic planarization layer (hereinafter “first OPL”) **152** may be formed on the structure **100**.

[0092] The first OPL **152** may be deposited and patterned on the structure **100**. The first OPL **152** may protect select sacrificial gates **122**, the gate hard mask **124**, the spacer **130**, the undoped silicon epitaxy **140**, the etch stop liner **150** and the STI **120** surrounding the sacrificial gates **122** along section line X2-X2.

[0093] The first OPL **152** may be formed by a blanket deposition using typical deposition techniques, for example spin-on coating. The first OPL **152** can be a self-planarizing organic material that includes carbon, hydrogen, oxygen, and optionally nitrogen, fluorine, and silicon. The first OPL **152** can be a standard C_xH_y polymer. Non-limiting examples of materials include, but are not limited to, CHM701B, commercially available from Cheil Chemical Co., Ltd., HM8006 and HM8014, commercially available from JSR Corporation, and ODL-102 or ODL-401, commercially available from ShinEtsu Chemical, Co., Ltd.

[0094] The first OPL **152** may be referred to as a first trench contact patterning mask. The first OPL **152** may provide access for subsequent patterning of the sacrificial gates **122** and surrounding materials along section line X1-X1 which are not protected by the first OPL **152**.

[0095] Referring now to FIGS. **35**, **36** and **37**, the structure **100** is shown according to an exemplary embodiment. FIGS.

35 and **36** are each a cross-sectional view of the structure **100** along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. **37** is a cross-sectional view of the structure **100** along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. A portion of the etch stop liner **150** and a portion of the undoped silicon epitaxy **140** may be removed between sacrificial gates **122** along section line X1-X1, forming a p-FET source drain trench **156**.

[0096] The p-FET source drain trench **156** may be formed between each spacer **130** surrounding each sacrificial gate **122** and gate hard mask **124** along section line X1-X1 by an anisotropic etching technique, such as, for example, reactive ion etching (RIE), removing a portion of the undoped silicon epitaxy **140**, where a remaining portion of the undoped silicon epitaxy **140** remains over the bottom dielectric isolation **36**. Removal of the portion of the etch stop liner **150** and the portion of the undoped silicon epitaxy **140** selective to the channel layers **116**, the sacrificial suspension layers **114**, the inner spacers **138**, the bottom dielectric isolation **146**, the seed layer **113** and the STI **120**. A vertical side surface may be exposed of each of the spacers **130** channel layers **116** and the inner spacers **138**. A portion of an upper surface of the STI **120** may be exposed.

[0097] The p-FET source drain trench **156** may be referred to as a self-aligned source drain epitaxy cut as the spacer **130** helps to protect the sacrificial gate **122**, the gate hard mask **124**, the channels layers **116** and the inner spacers **138**. The etch stop liner **150** patterning and removal does not damage inner spacer **138**.

[0098] Referring now to FIGS. **38**, **39** and **40**, the structure **100** is shown according to an exemplary embodiment. FIGS. **38** and **39** are each a cross-sectional view of the structure **100** along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. **40** is a cross-sectional view of the structure **100** along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. A p-FET source drain epitaxy **160** may be formed. The first OPL **152** may be removed.

[0099] The p-FET source drain epitaxy **160** may be epitaxially grown in a region on between adjacent sacrificial gates **122**, gate hard mask **124** and spacers **130**, in the p-FET source drain trench **156**. The p-FET source drain epitaxy **160** may be in direct contact with end portions of the channel layers **116** of the nanosheet stack and end portions of the inner spacer **138** surrounding the sacrificial layers **114**, along section line X1-X1. The p-FET source drain epitaxy **160** may be formed over the undoped silicon epitaxy **140**. The p-FET source drain epitaxy **160** may have two opposite side surfaces of the spacer **130**, and may have two additional opposite side surfaces of the nanosheet stack side surfaces of the inner spacer **138** and the channel layers **116**. An upper horizontal surface of the p-FET source drain epitaxy **160** may be above an upper surface of an uppermost layer of the nanosheet stacks.

[0100] The p-FET source drain epitaxy **160** is formed by epitaxial growth of a semiconductor material on physically exposed sidewalls of each channel layer **116** and from the undoped silicon epitaxy **140**.

[0101] A key benefit of this invention is the p-FET source drain epitaxy **160** can grow from both the exposed sidewalls of each channel layer **116** and from the undoped silicon epitaxy **140**, allowing for a high quality {100}-orientated

crystalline growth, allowing the generation of compressive strain on the channel layers **116** of a result p-FET.

[0102] The p-FET source drain epitaxy **160** includes any semiconductor material described above with respect to the substrate **102**. In some embodiments, the p-FET source drain epitaxy **160** may include the same semiconductor material as the channel layers **116**. In other embodiments, the p-FET source drain epitaxy **160** may include a different semiconductor material than that of the channel layers **116**. For example, the p-FET source drain epitaxy **160** may include a silicon germanium alloy, while the channel layers **116** may comprise silicon. The p-FET source drain epitaxy **160** may further be doped. Acceptable dopants may include either a p-type dopant or an n-type dopant. The term “p-type” refers to the addition of impurities to an intrinsic semiconductor that creates deficiencies of valence electrons. In a silicon-containing semiconductor material, examples of p-type dopants, include, but are not limited to, aluminum, beryllium, boron, cadmium, gallium, germanium, indium, silicon, and zinc. The term “n-type” refers to the addition of impurities that contributes free electrons to an intrinsic semiconductor. In a silicon containing semiconductor material, examples of n-type dopants include, but are not limited to, antimony, arsenic, germanium, phosphorous, selenium, silicon, and tellurium. In an embodiment, the p-FET source drain epitaxy **160** includes any semiconductor material described above is doped with p-type dopant for p-type FETs along the X1-X1 section.

[0103] In an embodiment, the p-type dopant may be heavily doped at levels of 5×10^{20} atoms/cm³ to 1×10^{21} atoms/cm³, although dopant levels may be high or lower and are dependent upon atoms used as dopants. In an embodiment, the n-type dopant may be heavily doped at levels of 1×10^{29} atoms/cm³ to 2×10^{29} atoms/cm³, although dopant levels may be high or lower and are dependent upon atoms used as dopants. In an embodiment, the undoped silicon epitaxy **140** may be doped at levels between about 1×10^{15} cm³ and 1×10^{16} atoms/cm³, although dopant levels may be high or lower and are dependent upon atoms used as dopants. A dopant level in the p-FET source drain epitaxy **160** may be at least 1000 times higher than that in the undoped silicon epitaxy **140**.

[0104] Referring now to FIGS. **41**, **42** and **43**, the structure **100** is shown according to an exemplary embodiment. FIGS. **41** and **41** are each a cross-sectional view of the structure **100** along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. **43** is a cross-sectional view of the structure **100** along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. An n-FET source drain epitaxy **164** may be formed.

[0105] Several steps similar to those described above may be performed. Another etch stop liner (not shown) may be conformed formed on the structure **100**, similar to the etch stop liner **150**. A second OPL (not shown), similar to the first OPL **152**, may be deposited and patterned on the structure **100**, along the X1-X1 section, surrounding sacrificial gates **122** and surrounding materials along section line X2-X2, such that sacrificial gates **122** and surrounding materials are not covered by the second OPL (not shown) along section line X1-X1. A portion of the another etch stop liner (not shown) and a portion of the undoped silicon epitaxy **140** may be removed between sacrificial gates **122** along section line X2-X2, forming an n-FET source drain trench (not shown), similar to the p-FET source drain trench **156**. The

n-FET source drain epitaxy **164** may be formed in the n-FET source drain trench (not shown), similar to the p-FET source drain epitaxy **160**. The second OPL (not shown) may be removed.

[0106] The n-FET source drain epitaxy **164** may be epitaxially grown in a region on between adjacent sacrificial gates **122**, gate hard mask **124** and spacers **130**, in the n-FET source drain trench (not shown). The n-FET source drain epitaxy **164** may be in direct contact with end portions of the channel layers **116** of the nanosheet stack and end portions of the inner spacer **138** surrounding the sacrificial layers **114**, along section line X2-X2. The n-FET source drain epitaxy **164** may be formed over the undoped silicon epitaxy **140**. The n-FET source drain epitaxy **164** may have two opposite side surfaces of the spacer **130**, and may have two additional opposite side surfaces of the nanosheet stack side surfaces of the inner spacer **138** and the channel layers **116**. An upper horizontal surface of the n-FET source drain epitaxy **164** may be above an upper surface of an uppermost layer of the nanosheet stacks.

[0107] The n-FET source drain epitaxy **164** is formed by epitaxial growth of a semiconductor material on physically exposed sidewalls of each channel layer **116** and from the undoped silicon epitaxy **140**.

[0108] A key benefit of this invention is the n-FET source drain epitaxy **160** can grow from both the exposed sidewalls of each channel layer **116** and from the undoped silicon epitaxy **140**, allowing for a high quality **100** crystalline growth, allowing for the generation of tensile strain on the nFET nanosheet channels.

[0109] The n-FET source drain epitaxy **164** includes any semiconductor material described above with respect to the substrate **100**. In some embodiments, the n-FET source drain epitaxy **164** may include the same semiconductor material as that of the channel layers **116**. In other embodiments, the n-FET source drain epitaxy **164** may include a different semiconductor material than that of the channel layers **116**. For example, the n-FET source drain epitaxy **164** may include SiC doped with phosphorous, while the channel layers **116** may comprise silicon. The n-FET source drain epitaxy **164** may further be doped. Acceptable dopants may include either a p-type dopant or an n-type dopant. The term “p-type” refers to the addition of impurities to an intrinsic semiconductor that creates deficiencies of valence electrons. In a silicon-containing semiconductor material, examples of p-type dopants, include, but are not limited to, aluminum, beryllium, boron, cadmium, gallium, germanium, indium, silicon, and zinc. The term “n-type” refers to the addition of impurities that contributes free electrons to an intrinsic semiconductor. In a silicon containing semiconductor material, examples of n-type dopants include, but are not limited to, antimony, arsenic, germanium, phosphorous, selenium, silicon, and tellurium.

[0110] In an embodiment, the n-FET source drain epitaxy **164** is doped with n-type dopant for n-type FETs along the X2-X2 section.

[0111] As described above, in an embodiment, the n-type dopant may be doped at levels of 1×10^{29} atoms/cm³ to 2×10^{29} atoms/cm³, although dopant levels may be high or lower and are dependent upon atoms used as dopants. In an embodiment, the undoped silicon epitaxy **140** may be doped at levels between about 1×10^{15} cm³ and 1×10^{16} atoms/cm³, although dopant levels may be high or lower and are dependent upon atoms used as dopants. A dopant level in the

n-FET source drain epitaxy **164** may be at least 1000 times higher than that in the undoped silicon epitaxy **140**.

[0112] In an embodiment, p-type FETs are made along section line X1-X1 and n-type FETs are made along section line X2-X2. Any combination of p-type FETs and n-type FETs may be made on the structure **100**.

[0113] Referring now to FIGS. **44**, **45** and **46**, the structure **100** is shown according to an exemplary embodiment. FIGS. **44** and **45** are each a cross-sectional view of the structure **100** along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. **46** is a cross-sectional view of the structure **100** along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. An interlayer dielectric ("ILD") **170** may be formed. The gate hard mask **124** and the sacrificial gate **122** may be removed. The sacrificial suspension layers **114** may be removed.

[0114] The ILD **170** may be formed after several processes, including for example, conformally depositing or growing a dielectric and performing an anisotropic etch process. The ILD **170** may include any dielectric material such as silicon nitride and may include a single layer or may include multiple layers of dielectric material. The ILD **170** may be formed between spacers **130** which are each on adjacent sacrificial gates **122**. The ILD **170** may be formed on the p-FET source drain epitaxy **160** and the n-FET source drain epitaxy **164**. The ILD **170** may cover an upper surface of the spacer **130**. The ILD **170** may have a vertical side surface of the spacer **130** and the bottom dielectric isolation **146**. The ILD **170** may be formed on the exposed portions of the STI **120**.

[0115] The structure **100** may be subject to a chemical mechanical polishing (CMP) technique to remove excess material, including the ILD **170** and the gate hard mask **124**, and polish upper surfaces of the structure **100**.

[0116] The sacrificial gate **122**, the gate hard mask **124** and the sacrificial suspension layers **114** may be removed by methods known in the art, and may be removed in one or more steps, selectively, or may be removed at the same time. For example, a dry etch process can be used, such as using vapor phased HCl dry etch. In an embodiment, an ammonia based etch, NH_3 , may be used for removal of the sacrificial gate **122**, a hydrogen fluoride based etch such as HF may be used to remove the gate hard mask **124**, and a hydrogen chloride, HCl, based etch may be used to remove the sacrificial suspension layers **114**. The sacrificial gate **122**, gate hard mask **124** and the sacrificial suspension layers **114** may be removed selective to the spacers **130**, the inner spacers **138**, the channel layers **116**, the bottom dielectric isolation **146**, the p-FET source drain epitaxy **160**, the n-FET source drain epitaxy **164**, the ILD **170** and the undoped silicon epitaxy **140**.

[0117] Referring now to FIGS. **47**, **48** and **49**, the structure **100** is shown according to an exemplary embodiment. FIGS. **47** and **48** are each a cross-sectional view of the structure **100** along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. **49** is a cross-sectional view of the structure **100** along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. A selective isotropic etch of the channel layers **116** may be performed and portions of the seed layer **113** may be removed.

[0118] A controlled oxidation of exposed portions of the channel layers **116** and of the seed layer **113** may be done by a wet or dry oxidation process providing sub-nanometer level of control, in areas below where the sacrificial gate **122**

and gate hard mask **124** were removed. Exposed portions of the inner spacers **138**, the channel layers **116** may not be affected by the controlled oxidation.

[0119] A selective isotropic etch of the exposed portions of the channel layers **116** may be done, resulting in a thinning of the oxidized portions of the channel layers **116** by 1 nanometer or less. This selective isotropic etch may result in a removal of the exposed portions of the seed layer **113**. The seed layer **113** may remain below the inner spacers **138** and below the undoped epitaxy **140**, and above the bottom dielectric isolation **146**.

[0120] Referring now to FIGS. **50**, **51** and **52**, the structure **100** is shown according to an exemplary embodiment. FIGS. **50** and **51** are each a cross-sectional view of the structure **100** along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. **52** is a cross-sectional view of the structure **100** along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2. A gate stack **176** may be formed.

[0121] The gate stack **176** may be formed in openings where the gate hard mask **124**, the sacrificial gate **122** and the sacrificial suspension layers **114** were removed, and where the exposed portion of the seed layer **113** was removed.

[0122] The gate stack **176** may be conformally formed on the structure **100**, according to an exemplary embodiment. The gate stack **176** is formed in each cavity of the nanosheet stack and surrounding suspended portions of the channel layers **116**. The gate stack **176** may be wrapped around each channel layer **116**.

[0123] The gate stack **176** forms a layer surrounding exposed portions of the nanosheet stacks. The gate stack **176** may cover an exposed portion of the STI **120**, between exposed portions of the bottom dielectric isolation **146** and a lowermost channel layer **116**, exposed surfaces between the inner spacers **138**, exposed portions surrounding the channel layers **116** and exposed surfaces between the spacers **130** over the nanosheet stack. The gate stack **176** may fill a remainder of openings where the sacrificial suspension layers **114** were removed and the portions of the seed layer **113** was removed. The gate stack **176** may fill an opening where the sacrificial gate **122** was removed between spacers **130** over the nanosheet stack.

[0124] In an embodiment, a high-k liner, not shown, may be conformally formed before formation of the gate stack **176**, and the gate stack **176** formed on the high-k liner. The high-k liner, not shown, may cover an exposed portion of the STI **120**, surround exposed portions of the bottom dielectric isolation **146**, exposed surfaces between the inner spacers **138** and exposed surfaces between the spacers **130** over the nanosheet stack. The high-k liner, not shown, may be deposited using typical deposition techniques, for example, atomic layer deposition (ALD), molecular layer deposition (MLD), chemical vapor deposition (CVD), physical vapor deposition (PVD), high density plasma (HDP) deposition, and spin on techniques, followed by an anisotropic vertical etch process such as a reactive ion etch (RIE), or any suitable etch process. In an embodiment, the high-k liner, not shown, may include one or more layers. The high-k liner, not shown, may include materials such as HfO_2 , ZrO_2 , Al_2O_3 , La_2O_3 , TiO_2 , SrTiO_3 , LaAlO_3 , Y_2O_3 , HfOxNy , ZrOxNy , La_2OxNy , Al_2OxNy , TiOxNy , SrTiOxNy , LaAlOxNy , Y_2OxNy , SiON , SiNx , a silicate thereof, and an alloy thereof. The high-k liner, not shown, may be referred to as

a gate dielectric, and may be 2 nm thick. The high-k liner, not shown, may serve as the gate oxide for future gate formation.

[0125] The gate stack 176 may be deposited using typical deposition techniques, for example, atomic layer deposition (ALD), molecular layer deposition (MLD), and chemical vapor deposition (CVD). The material chosen for the gate stack 176, and the high-k liner, not shown, may be selected based on a desired threshold voltage, in combination with other materials and properties as described above, for the nanosheet stack, where the gate stack 176 surrounds the channel layers 116, and whether the device is a p-FET or n-FET. In an embodiment, the work function metal of a p-FET device may include a metal nitride, for example, titanium nitride or tantalum nitride, titanium carbide titanium aluminum carbide, or other suitable materials known in the art. In an embodiment, the work function metal of an n-FET device may include, for example, titanium aluminum carbide or other suitable materials known in the art. In an embodiment, the gate stack 176 may include one or more layers to achieve desired device characteristics.

[0126] After forming the gate stack 176, a chemical mechanical polishing (CMP) technique may be used to remove excess material and polish upper surfaces of the structure 100 such that upper horizontal surfaces of the gate stack 176, the spacers 130 and the ILD 170 are coplanar.

[0127] The resulting structure has a benefit of the p-FET source drain epitaxy 160 and the n-FET source drain epitaxy 164 growing with strain from both the undoped silicon epitaxy 140 and the channel layers 116 of the nanosheet stack.

[0128] In a fully integrated device, a dielectric may cover the structure 100, and contacts may be made to each gate and to each of the p-FET source drain epitaxys 160, and the n-FET source drain epitaxys 164.

[0129] Referring now to FIGS. 53, 54 and 55, the structure 100 is shown according to an alternate embodiment. FIGS. 53 and 54 are each a cross-sectional view of the structure 100 along section lines X1-X1 and X2-X2, respectively, and are parallel to each other. FIG. 55 is a cross-sectional view of the structure 100 along section line Y-Y and is perpendicular to section lines X1-X1 and X2-X2.

[0130] In comparison of FIGS. 53, 54 and 55 to FIGS. 50, 51 and 52, along section line X1-X1, the bottom dielectric isolation 146 is replaced by a second bottom dielectric isolation 180. The bottom dielectric isolation 146 remains along section line X2-X2. Additionally, along section line X1-X1, the p-FET source drain epitaxy 160 has a lower horizontal surface along an upper surface of the second bottom dielectric isolation 180, rather than having the undoped silicon epitaxy 140 between the p-FET source drain epitaxy 160 and the bottom dielectric isolation 146.

[0131] Changes in the process described above include during the steps described in FIGS. 26, 27 and 28, rather than forming the bottom dielectric isolation 146 uniformly in the gap 144, sections line X1-X1 and X2-X2 would be separately processed, with the bottom dielectric isolation 146 formed along section line X2-X2, and the second bottom dielectric isolation 180 formed along section line X1-X1. Furthermore, for the steps described in FIGS. 35, 36, 37, rather than removing a portion of the undoped silicon epitaxy 140 between adjacent sacrificial gates 122, the

undoped silicon epitaxy 140 may be completely removed, exposing an upper surface of what is now the second bottom dielectric isolation 180.

[0132] In this embodiment, the second bottom dielectric 180 may be formed of the same material as the spacer 130.

[0133] The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the invention. The terminology used herein was chosen to best explain the principles of the embodiment, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

What is claimed is:

1. A semiconductor gate-all-around device comprising: semiconductor channel layers vertically aligned and stacked one on top of another, the semiconductor channel layers separated from each other by a gate stack material wrapping around the semiconductor channel layers; and
 - a heavily doped p-type field effect transistor (p-FET) source drain epitaxy region adjacent to the semiconductor channel layers, wherein a horizontal lower surface of the p-FET source drain epitaxy region is adjacent to a horizontal upper surface of an undoped silicon epitaxy.
2. The semiconductor gate-all-around device according to claim 1, further comprising:
 - a bottom isolation region aligned below the undoped silicon epitaxy and below the semiconductor channel layers.
3. The semiconductor gate-all-around device according to claim 1, further comprising:
 - a heavily doped n-type field effect transistor (n-FET) source drain epitaxy doped region adjacent to a second set of semiconductor channel layers, wherein a horizontal lower surface of the n-FET source-drain epitaxy region is adjacent to the horizontal upper surface of the undoped silicon epitaxy.
4. The semiconductor gate-all-around device according to claim 1, wherein
 - an upper horizontal surface of the undoped silicon epitaxy is below a lower horizontal surface of a lowermost channel layer of the semiconductor channel layers.
5. The semiconductor gate-all-around device according to claim 1, further comprising:
 - an inner spacer surrounding the gate stack material, the gate stack material wrapping around the semiconductor channel layers, and adjacent to the heavily doped source drain epitaxy region.
6. The semiconductor gate-all-around device according to claim 1, wherein
 - each of semiconductor channel layers comprise a height wider along an outer portion of each semiconductor channel layer and a height thinner along a center portion of each channel layer, the center portion aligned with the gate stack material.
7. A semiconductor device comprising:
 - a first set of semiconductor channel layers vertically aligned and stacked one on top of another;

- a heavily doped p-type field effect transistor (p-FET) source drain epitaxy region adjacent to the first set of semiconductor channel layers, wherein a horizontal lower surface of the p-FET source drain epitaxy region is adjacent to a horizontal upper surface of an undoped silicon epitaxy; and
 - a heavily doped n-type field effect transistor (n-FET) source drain epitaxy doped region adjacent to a second set of semiconductor channel layers, wherein a horizontal lower surface of the n-FET source-drain epitaxy region is adjacent to the horizontal upper surface of the undoped silicon epitaxy.
8. The semiconductor device according to claim 7, further comprising:
- a bottom isolation region aligned below the un-doped silicon epitaxy and below the semiconductor channel layers.
9. The semiconductor device according to claim 7, wherein
- an upper horizontal surface of the undoped silicon epitaxy is below a lower horizontal surface of a lowest channel layer of the semiconductor channel layers.
10. The semiconductor device according to claim 7, wherein
- an inner spacer surrounding gate stack material, the gate stack material wrapping around the semiconductor channel layers, and adjacent to the heavily doped p-FET source drain epitaxy region.
11. The semiconductor device according to claim 10, wherein
- each of semiconductor channel layers comprise a height wider along an outer portion of each semiconductor channel layer and a height thinner along a center portion of each channel layer, the center portion aligned with the gate stack material.
12. The semiconductor device according to claim 7, wherein
- a lower surface of a conductive trench contact is adjacent to an upper surface of a portion of the source drain epitaxy region.
13. A method comprising:
- forming a first stack of nanosheet layers on a substrate, a second stack of nanosheet layers on the substrate, and a third stack of nanosheet layers on the substrate, the first stack, the second stack and the third stack of nanosheet layers each comprising alternating layers of a sacrificial and a semiconductor channel vertically aligned and stacked one on top of another;
 - forming a first sacrificial gate across the first stack of nanosheet layers, a second sacrificial gate across the second stack of nanosheet layers and a third sacrificial gate across the third stack of nanosheet layers; and
 - forming an undoped silicon epitaxy between the first stack and the second stack of nanosheet layers and between the second stack and the third stack of nanosheet layers.
14. The method according to claim 13, further comprising:
- after forming the undoped silicon epitaxy between the first stack and the second stack of nanosheet layers and between the second stack and the third stack of nanosheet layers,
 - replacing a sacrificial epitaxial layer under the first stack, second stack and third stack of nanosheet layers and under the undoped silicon epitaxy, with a continuous bottom dielectric isolation.
15. The method according to claim 14, further comprising:
- replacing a portion of the undoped silicon epitaxy between the first stack and the second stack of nanosheet layers with a heavily p-type field effect transistor (p-FET) source drain epitaxy region; and
 - replacing a portion of the undoped silicon epitaxy between the second stack and the third stack of nanosheet layers with a heavily n-type field effect transistor (n-FET) source drain epitaxy region.
16. The method according to claim 13, further comprising:
- replacing the first, second and third sacrificial gate with a first, second and third gate stack.
17. The method according to claim 13, further comprising:
- after forming the undoped silicon epitaxy between the first stack and the second stack of nanosheet layers and between the second stack and the third stack of nanosheet layers,
 - replacing a sacrificial epitaxial layer under the first stack and second stack of nanosheet layers and under the undoped silicon epitaxy between the first stack and the second stack of nanosheet layers with a first continuous bottom dielectric isolation; and
 - replacing a sacrificial epitaxial layer under the third stack of nanosheet layers and under the undoped silicon epitaxy between the second stack and the third stack of nanosheet layers with a second continuous bottom dielectric isolation.
18. The method according to claim 17, further comprising:
- replacing the undoped silicon epitaxy between the first stack and the second stack of nanosheet layers with a heavily p-type field effect transistor (p-FET) source drain epitaxy region; and
 - replacing a portion of the undoped silicon epitaxy between the second stack and the third stack of nanosheet layers with a heavily n-type field effect transistor (n-FET) source drain epitaxy region.
19. The method according to claim 13, wherein
- an upper horizontal surface of the undoped silicon epitaxy is below a lower horizontal surface of a lowest channel layer of the first stack of nanosheet layers.
20. The method according to claim 13, wherein
- an inner spacer surrounding gate stack material, the gate stack material wrapping around the semiconductor channel layers of the first stack of nanosheet layers, and adjacent to the heavily doped p-FET source drain epitaxy region.